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Dual Nature of Radiation

JEE Main Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 197 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

A monochromatic source of light operating at 15 kW emits 2.5×10^{22} photons $/\mathrm{s}$. The region of an electromagnetic spectrum to which the emitted electromagnetic radiation belongs to ____.

(Take $h = 6.6 \times 10^{-34}\ \mathrm{J.s}$ and $c = 3 \times 10^{8}\ \mathrm{m/s}$).

(A) Microwave (B) Infrared (C) Visible (D) Ultraviolet

Open solution

Q2. JEE Main 2026 (Online) 8th April Evening Shift

K_1 and K_2 be the maximum kinetic energies of photoelectrons emitted from a surface of a given material for the light of wavelength λ_1 and λ_2, respectively. If $\lambda_1 = 2\lambda_2$ then the work function of material is given by :

(A) $K_2 + 2K_1$ (B) $2K_2 - K_1$ (C) $K_1 - 2K_2$ (D) $K_2 - 2K_1$

Open solution

Q3. JEE Main 2026 (Online) 5th April Evening Shift

An electron is travelling with a velocity v in free space and when it enters a medium, its velocity is reduced by 20%. The de Broglie wavelength of electron in the medium is $\alpha\lambda_0$, where λ_0 is its de Broglie wavelength in free space. The value of α is ____ .

(A) 1.20 (B) 1.0 (C) 1.25 (D) 0.75

Open solution

Q4. JEE Main 2026 (Online) 5th April Morning Shift

An electron of mass m is moving in an electric field $\vec{E} = -2E_o\hat{i}\ (E_o = \text{constant} > 0)$, with an initial velocity $\vec{V} = v_o\hat{i}\ (v_o = \text{constant} > 0)$. If $\lambda_o = \frac{h}{4mv_o}$, its de Broglie wavelength at time t is ____ .

($e =$ charge of electron)

(A) $\frac{4\lambda_o}{\left[1-\frac{E_o e}{2m}\frac{t}{v_o}\right]}$

(B) $\frac{4\lambda_o}{\left[1+\frac{E_o e}{2m}\frac{t}{v_o}\right]}$

(C) $\frac{4\lambda_o}{\left[1+\frac{2E_o e}{m}\frac{t}{v_o}\right]}$

(D) $\frac{4\lambda_o}{\left[1-\frac{2E_o e}{m}\frac{t}{v_o}\right]}$

Open solution

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Q5. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

Light source having wavelength 331 nm is used to generate photo-electrons whose stopping potential is 0.2 V . The work function of the used metal in the experiment is $\alpha \times 10^{-19}$ J. The value of α is ____ .

$\left(\mathrm{h} = 6.62 \times 10^{-34}\ \mathrm{J\,s}, \mathrm{e} = 1.6 \times 10^{-19}\mathrm{C}\ \text{and}\ \mathrm{c} = 3 \times 10^{8}\ \mathrm{m/s}\right)$

(A) 3.68 (B) 4.68 (C) 5.68 (D) 2.68

Q6. JEE Main 2026 (Online) 4th April Evening Shift

Open solution

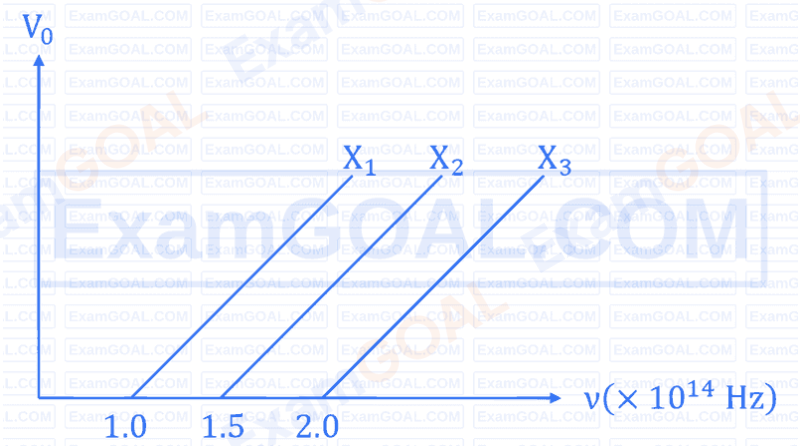
The de Broglie wavelength associated with an electron accelerated through a potential difference V is λ_e and the de Broglie wavelength associated with a proton accelerated through the same potential difference is λ_p. If their corresponding masses are m_e and m_p, respectively, then the ratio of their de Broglie wavelengths $\left(\frac{\lambda_e}{\lambda_p}\right)$ is ____ .

(A) $\sqrt{\frac{m_p}{m_e}}$ (B) $\sqrt{\frac{m_e}{m_p}}$ (C) $\frac{m_p}{m_e}$ (D) $\left(\frac{m_p}{m_e}\right)^2$

Q7. JEE Main 2026 (Online) 4th April Morning Shift

Open solution

The graph shows variation of stopping potential V_o with the frequency v of the incident radiation for three photosensitive metals X_1, X_2 and X_3. Which metal will give out electrons with greater kinetic energy, for the same wavelength of incident radiation?



(A) X_1

(B) X_2

(C) X_3

(D) All the metals will give out photo electrons with same kinetic energies.

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Q8. JEE Main 2026 (Online) 2nd April Morning Shift

For a certain metal, when monochromatic light of wavelength λ is incident, the stopping potential for photoelectrons is $3V_0$. When the same metal is illuminated by light of wavelength 2λ, then the stopping potential becomes V_0. The threshold wavelength for photoelectric emission for the given metal is $\alpha\lambda$. The value of α is _____.

Open solution

(A) 1 (B) 4 (C) 2 (D) 3

Q9. JEE Main 2026 (Online) 28th January Evening Shift

Number of photons of equal energy emitted per second by a 6 mW laser source operating at 663 nm is _____.

(Given : $h = 6.63 \times 10^{-34}$ J.s and $c = 3 \times 10^8$ m/s)

Open solution

(A) 10×10^{15} (B) 5×10^{16} (C) 5×10^{15} (D) 2×10^{16}

Q10. JEE Main 2026 (Online) 24th January Evening Shift

When a light of a given wavelength falls on a metallic surface the stopping potential for photoelectrons is 3.2 V . If a second light having wavelength twice of first light is used, the stopping potential drops to 0.7 V . The wavelength of first light is ____ m .

$\left(\mathrm{h} = 6.63 \times 10^{-34}\ \mathrm{J.s}, \mathrm{e} = 1.6 \times 10^{-19}\mathrm{C}, \mathrm{c} = 3 \times 10^8\ \mathrm{m/s}\right)$

Open solution

(A) 2.9×10^{-8} (B) 2.5×10^{-7} (C) 3.1×10^{-7} (D) 2.2×10^{-8}

Q11. JEE Main 2026 (Online) 23rd January Morning Shift

The de Broglie wavelength of an oxygen molecule at $27^{\circ}\mathrm{C}$ is $x \times 10^{-12}$ m. The value of x is (take Planck's constant $= 6.63 \times 10^{-34}$ J. s, Boltzmann constant $= 1.38 \times 10^{-23}$ J/K, mass of oxygen molecule $= 5.31 \times 10^{-26}$ kg)

Open solution

(A) 24 (B) 30 (C) 20 (D) 26

Q12. JEE Main 2026 (Online) 22nd January Evening Shift

Light is incident on a metallic plate having work function 110×10^{-20} J. If the produced photoelectrons have zero kinetic energy then the angular frequency of the incident light is ____ rad/s. $\left(\mathrm{h} = 6.63 \times 10^{-34}\ \mathrm{J.s}\right)$.

Open solution

(A) 1.04×10^{13} (B) 1.66×10^{16} (C) 1.66×10^{15} (D) 1.04×10^{16}

Q13. JEE Main 2026 (Online) 21st January Morning Shift

A light wave described by $E = 60\left[\sin\left(3 \times 10^{15}\right)t + \sin\left(12 \times 10^{15}\right)t\right]$ (in SI units) falls on a metal surface of work function 2.8 eV . The maximum kinetic energy of ejected photoelectron is (approximately)

____ eV. $\left(h = 6.6 \times 10^{-34}\ \text{J.s. and } e = 1.6 \times 10^{-19}\mathrm{C}\right)$

Open solution

(A) 3.8 (B) 7.8 (C) 6.0 (D) 5.1

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Q14. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

A photoemissive substance is illuminated with a radiation of wavelength λ_i so that it releases electrons with de-Broglie wavelength λ_e. The longest wavelength of radiation that can emit photoelectron is λ_o. Expression for de-Broglie wavelength is given by:

(m: mass of the electron, h: Planck's constant and c: speed of light)

(A) $\lambda_e = \frac{\sqrt{h\lambda_i}}{\sqrt{2mc}}$

(B) $\lambda_e = \frac{h}{\sqrt{2mc\left(\frac{1}{\lambda_i}-\frac{1}{\lambda_o}\right)}}$

(C) $\lambda_e = \sqrt{\frac{h}{2mc\left(\frac{1}{\lambda_i}-\frac{1}{\lambda_o}\right)}}$

(D) $\lambda_e = \sqrt{\frac{h\lambda_0}{2mc}}$

Q15. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

A small mirror of mass m is suspended by a massless thread of length l. Then the small angle through which the thread will be deflected when a short pulse of laser of energy E falls normal on the mirror

(c = speed of light in vacuum and g = acceleration due to gravity)

(A) $\theta = \frac{E}{mc\sqrt{gl}}$

(B) $\theta = \frac{E}{2mc\sqrt{gl}}$

(C) $\theta = \frac{3E}{4mc\sqrt{gl}}$

(D) $\theta = \frac{2E}{mc\sqrt{gl}}$

Q16. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A : In photoelectric effect, on increasing the intensity of incident light the stopping potential increases.

Reason R : Increase in intensity of light increases the rate of photoelectrons emitted, provided the frequency of incident light is greater than threshold frequency.

In the light of the above statements, choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(C) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(D) $\mathbf{A}$ is true but $\mathbf{R}$ is false

Q17. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

The work function of a metal is 3 eV . The color of the visible light that is required to cause emission of photoelectrons is

(A) Red (B) Green (C) Blue (D) Yellow

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Q18. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

The radiation pressure exerted by a 450 W light source on a perfectly reflecting surface placed at 2 m away from it, is

(A) 3×10^{-8} Pascals

(B) 0

(C) 1.5×10^{-8} Pascals

(D) 6×10^{-8} Pascals

Q19. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

An electron with mass ' m ' with an initial velocity $(\mathrm{t}=0)\overrightarrow{\mathrm{v}}=\mathrm{v}_0\hat{i}\,(\mathrm{v}_0>0)$ enters a magnetic field $\overrightarrow{\mathrm{B}}=\mathrm{B}_0\hat{j}$. If the initial de-Broglie wavelength at $\mathrm{t}=0$ is λ_0 then its value after time ' t ' would be :

(A) $\frac{\lambda_0}{\sqrt{1-\frac{\mathrm{e}^2\mathrm{B}_0^2\mathrm{t}^2}{\mathrm{m}^2}}}$

(B) λ_0

(C) $\lambda_0\sqrt{1+\frac{\mathrm{e}^2\mathrm{B}_0^2\mathrm{t}^2}{\mathrm{m}^2}}$

(D) $\frac{\lambda_0}{\sqrt{1+\frac{\mathrm{e}^2\mathrm{B}_0^2\mathrm{t}^2}{\mathrm{m}^2}}}$

Q20. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

A monochromatic light is incident on a metallic plate having work function ϕ. An electron, emitted normally to the plate from a point A with maximum kinetic energy, enters a constant magnetic field, perpendicular to the initial velocity of electron. The electron passes through a curve and hits back the plate at a point B. The distance between A and B is : (Given : The magnitude of charge of an electron is e and mass is m, h is Planck's constant and c is velocity of light. Take the magnetic field exists throughout the path of electron)

(A) $\sqrt{2m\left(\frac{hc}{\lambda}-\phi\right)}/\mathrm{eB}$

(B) $\sqrt{8\,\mathrm{m}\left(\frac{\mathrm{hc}}{\lambda}-\phi\right)}/\mathrm{eB}$

(C) $\sqrt{\mathrm{m}(\mathrm{hc}/\lambda-\phi)}/\mathrm{eB}$

(D) $2\sqrt{\mathrm{m}(\mathrm{hc}/\lambda-\phi)}/\mathrm{eB}$

Q21. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

In an experiment with photoelectric effect, the stopping potential,

(A) is $\left(\frac{1}{e}\right)$ times the maximum kinetic energy of the emitted photoelectrons

(B) increases with increase in the intensity of the incident light

(C) decreases with increase in the intensity of the incident light

(D) increases with increase in the wavelength of the incident light

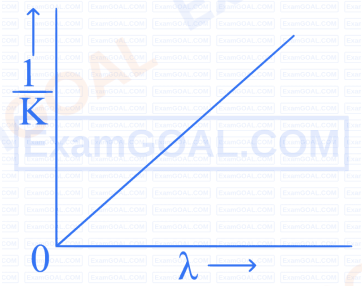
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Q22. JEE Main 2025 (Online) 29th January Morning Shift

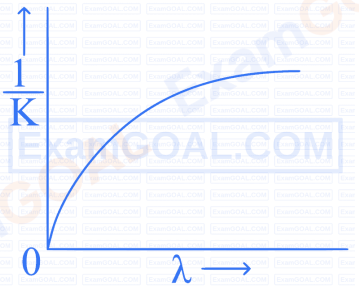
If λ and K are de Broglie wavelength and kinetic energy, respectively, of a particle with constant mass. The correct graphical representation for the particle will be :

Open solution

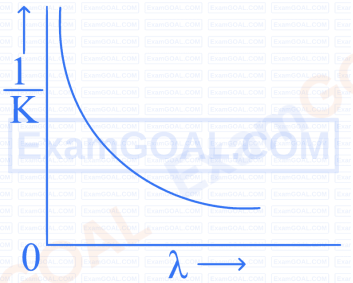
(A)



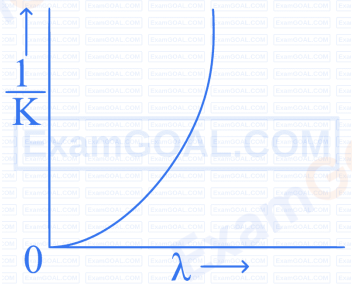
(B)



(C)



(D)



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Q23. JEE Main 2025 (Online) 29th January Morning Shift

Open solution

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Emission of electrons in photoelectric effect can be suppressed by applying a sufficiently negative electron potential to the photoemissive substance.

Reason (R) : A negative electric potential, which stops the emission of electrons from the surface of a photoemissive substance, varies linearly with frequency of incident radiation.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is false but (R) is true

(D) (A) is true but (R) is false

Q24. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

Which of the following phenomena cannot be explained by wave theory of light?

(A) Refraction of light

(B) Reflection of light

(C) Diffraction of light

(D) Compton effect

Q25. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

A proton of mass ' m_P ' has same energy ' E ' as that of a photon of wavelength ' λ '. If the proton is moving at non-relativistic speed, then ratio of its de Broglie wavelength to the wavelength of photon is.

(A) $\frac{1}{c}\sqrt{\frac{E}{m_p}}$

(B) $\frac{1}{\mathrm{c}}\sqrt{\frac{2\mathrm{E}}{\mathrm{m_p}}}$

(C) $\frac{1}{2\mathrm{c}}\sqrt{\frac{\mathrm{E}}{\mathrm{m_p}}}$

(D) $\frac{1}{\mathrm{c}}\sqrt{\frac{\mathrm{E}}{2\,\mathrm{m_p}}}$

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Q26. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

In photoelectric effect, the stopping potential $(\mathrm{V_0})\mathrm{v/s}$ frequency (ν) curve is plotted.

(h is the Planck's constant and ϕ_0 is work function of metal)

(A) $\mathrm{V_0 v/s}\ \nu$ is linear.

(B) The slope of $\mathrm{V_0 v/s}\ \nu$ curve $= \frac{\phi_0}{\mathrm{h}}$

(C) h constant is related to the slope of $\mathrm{V_0 v/s}\ \nu$ line.

(D) The value of electric charge of electron is not required to determine h using the $\mathrm{V_0 v/s}\ \nu$ curve.

(E) The work function can be estimated without knowing the value of h.

Choose the correct answer from the options given below :

(A) (A), (C) and (E) only

(B) (C) and (D) only

(C) (A), (B) and (C) only

(D) (D) and (E) only

Q27. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

An electron of mass ' m ' with an initial velocity $\overrightarrow{\mathrm{v}} = \mathrm{v_0}\hat{i}\,(\mathrm{v_0} > 0)$ enters an electric field $\overrightarrow{\mathrm{E}} = -\mathrm{E_o}\hat{\mathrm{k}}$. If the initial de Broglie wavelength is λ_0, the value after time t would be

(A) $\frac{\lambda_o}{\sqrt{1-\frac{\mathrm{e^2E_o^2t^2}}{\mathrm{m^2v_o^2}}}}$

(B) λ_0

(C) $\frac{\lambda_o}{\sqrt{1+\frac{\mathrm{e^2E_o^2t^2}}{\mathrm{m^2}v_o^2}}}$

(D) $\lambda_\mathrm{o}\sqrt{1+\frac{\mathrm{e^2E_o^2t^2}}{\mathrm{m^2v_o^2}}}$

Q28. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

In photoelectric effect an em-wave is incident on a metal surface and electrons are ejected from the surface. If the work function of the metal is 2.14 eV and stopping potential is 2 V , what is the wavelength of the em-wave? (Given $\mathrm{hc} = 1242\mathrm{eVnm}$ where h is the Planck's constant and c is the speed of light in vacuum.)

(A) 400 nm **(B)** 600 nm **(C)** 300 nm **(D)** 200 nm

Q29. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

A sub-atomic particle of mass $10^{-30}\ \mathrm{kg}$ is moving with a velocity $2.21 \times 10^6\ \mathrm{m/s}$. Under the matter wave consideration, the particle will behave closely like $\left(\mathrm{h} = 6.63 \times 10^{-34}\ \mathrm{J.s}\right)$

(A) X-rays

(B) Infra-red radiation

(C) Gamma rays

(D) Visible radiation

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Q30. JEE Main 2025 (Online) 22nd January Evening Shift

A light source of wavelength λ illuminates a metal surface and electrons are ejected with maximum kinetic energy of 2 eV . If the same surface is illuminated by a light source of wavelength $\frac{\lambda}{2}$, then the maximum kinetic energy of ejected electrons will be (The work function of metal is 1 eV)

(A) 5 eV (B) 3 eV (C) 2 eV (D) 6 eV

Open solution

Q31. JEE Main 2025 (Online) 22nd January Morning Shift

The work functions of cesium (Cs) and lithium (Li) metals are 1.9 eV and 2.5 eV , respectively. If we incident a light of wavelength 550 nm on these two metal surfaces, then photo-electric effect is possible for the case of

(A) Both Cs and Li (B) Neither Cs nor Li

(C) Li only (D) Cs only

Open solution

Q32. JEE Main 2025 (Online) 22nd January Morning Shift

An electron in the ground state of the hydrogen atom has the orbital radius of $5.3 \times 10^{-11}\ \mathrm{m}$ while that for the electron in third excited state is $8.48 \times 10^{-10}\ \mathrm{m}$. The ratio of the de Broglie wavelengths of electron in the excited state to that in the ground state is

(A) 4 (B) 3 (C) 9 (D) 16

Open solution

Q33. JEE Main 2024 (Online) 9th April Evening Shift

UV light of $4.13\ \mathrm{eV}$ is incident on a photosensitive metal surface having work function $3.13\ \mathrm{eV}$. The maximum kinetic energy of ejected photoelectrons will be:

(A) 4.13 eV (B) 1 eV (C) 7.26 eV (D) 3.13 eV

Open solution

Q34. JEE Main 2024 (Online) 9th April Morning Shift

A proton, an electron and an alpha particle have the same energies. Their de-Broglie wavelengths will be compared as :

(A) $\lambda_{\mathrm{p}} > \lambda_{\mathrm{e}} > \lambda_{\alpha}$ (B) $\lambda_{\alpha} < \lambda_{\mathrm{p}} < \lambda_{\mathrm{e}}$

(C) $\lambda_{\mathrm{e}} > \lambda_{\alpha} > \lambda_{\mathrm{p}}$ (D) $\lambda_{\mathrm{p}} < \lambda_{\mathrm{e}} < \lambda_{\alpha}$

Open solution

Q35. JEE Main 2024 (Online) 8th April Evening Shift

A proton and an electron have the same de Broglie wavelength. If $\mathrm{K_p}$ and $\mathrm{K_e}$ be the kinetic energies of proton and electron respectively, then choose the correct relation :

(A) $\mathrm{K_p} > \mathrm{K_e}$ (B) $\mathrm{K_p} = \mathrm{K_e}$

(C) $\mathrm{K_p} < \mathrm{K_e}$ (D) $\mathrm{K_p} = \mathrm{K_e}^2$

Open solution

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Q36. JEE Main 2024 (Online) 8th April Morning Shift

A proton and an electron are associated with same de-Broglie wavelength. The ratio of their kinetic energies is:

(Assume h = 6.63 $\times 10^{-34}$ J s, $m_e = 9.0 \times 10^{-31}$ kg and $m_p = 1836$ times m_e)

(A) $1 : \frac{1}{1836}$ (B) $1 : \sqrt{1836}$ (C) $1 : 1836$ (D) $1 : \frac{1}{\sqrt{1836}}$

Open solution

Q37. JEE Main 2024 (Online) 6th April Evening Shift

When UV light of wavelength 300 nm is incident on the metal surface having work function 2.13 eV, electron emission takes place. The stopping potential is :

(Given hc $= 1240$ eV nm)

(A) 4 V (B) 2 V (C) 4.1 V (D) 1.5 V

Open solution

Q38. JEE Main 2024 (Online) 6th April Morning Shift

Which of the following phenomena is not explained by wave nature of light?

A. reflection

B. diffraction

C. photoelectric effect

D. interference

E. polarization

Choose the most appropriate answer from the options given below:

(A) C only (B) B, D only (C) A, C only (D) E only

Open solution

Q39. JEE Main 2024 (Online) 6th April Morning Shift

In photoelectric experiment energy of 2.48 eV irradiates a photo sensitive material. The stopping potential was measured to be 0.5 V. Work function of the photo sensitive material is :

(A) 1.98 eV (B) 1.68 eV (C) 2.48 eV (D) 0.5 eV

Open solution

Q40. JEE Main 2024 (Online) 5th April Evening Shift

Which of the following statement is not true about stopping potential (V_0) ?

(A) It depends upon frequency of the incident light.

(B) It is $1/e$ times the maximum kinetic energy of electrons emitted.

(C) It increases with increase in intensity of the incident light.

(D) It depends on the nature of emitter material.

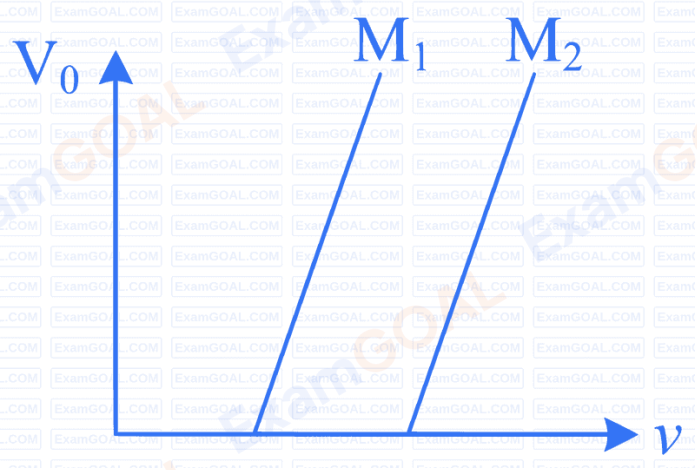
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Q41. JEE Main 2024 (Online) 5th April Morning Shift

Open solution

Given below are two statements :



Statement I : Figure shows the variation of stopping potential with frequency (v) for the two photosensitive materials M_1 and M_2. The slope gives value of $\frac{h}{e}$, where h is Planck's constant, e is the charge of electron.

Statement II : M_2 will emit photoelectrons of greater kinetic energy for the incident radiation having same frequency.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Statement I is correct and Statement II is incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are correct

(D) Both Statement I and Statement II are incorrect

Q42. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason R.

Assertion A: Number of photons increases with increase in frequency of light.

Reason R: Maximum kinetic energy of emitted electrons increases with the frequency of incident radiation.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) $\mathbf{A}$ is not correct but $\mathbf{R}$ is correct.

(B) $\mathbf{A}$ is correct but $\mathbf{R}$ is not correct.

(C) Both $\mathbf{A}$ and $\mathbf{R}$ are correct and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$.

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are correct and $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$.

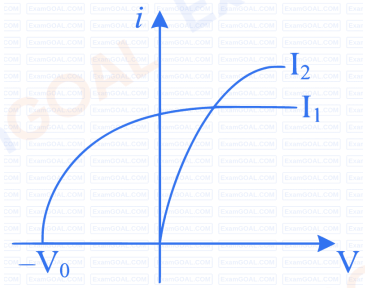
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Q43. JEE Main 2024 (Online) 4th April Morning Shift

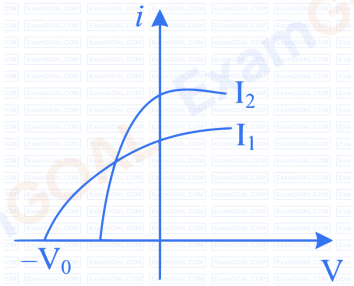
Open solution

Which figure shows the correct variation of applied potential difference (V) with photoelectric current (I) at two different intensities of light $(I_1 < I_2)$ of same wavelengths :

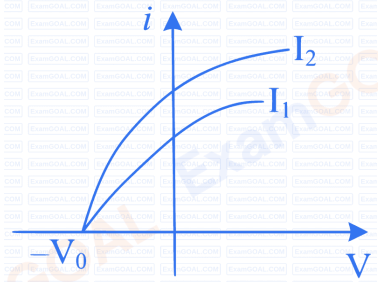
(A)



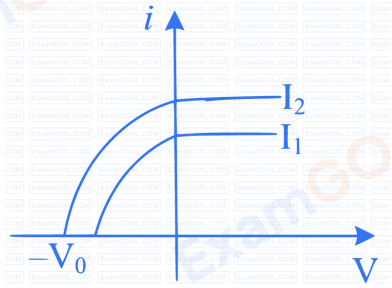
(B)



(C)



(D)



Q44. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

Monochromatic light of frequency 6×10^{14} Hz is produced by a laser. The power emitted is 2×10^{-3} W.

How many photons per second on an average, are emitted by the source ?

(Given $h = 6.63 \times 10^{-34}$Js)

(A) 5×10^{15} (B) 7×10^{16} (C) 6×10^{15} (D) 9×10^{18}

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Q45. JEE Main 2024 (Online) 1st February Morning Shift

The de Broglie wavelengths of a proton and an α particle are λ and 2λ respectively. The ratio of the velocities of proton and α particle will be :

(A) $8:1$ (B) $1:2$ (C) $1:8$ (D) $4:1$

Open solution

Q46. JEE Main 2024 (Online) 31st January Evening Shift

In a photoelectric effect experiment a light of frequency 1.5 times the threshold frequency is made to fall on the surface of photosensitive material. Now if the frequency is halved and intensity is doubled, the number of photo electrons emitted will be:

(A) doubled (B) halved (C) Zero (D) quadrupled

Open solution

Q47. JEE Main 2024 (Online) 31st January Morning Shift

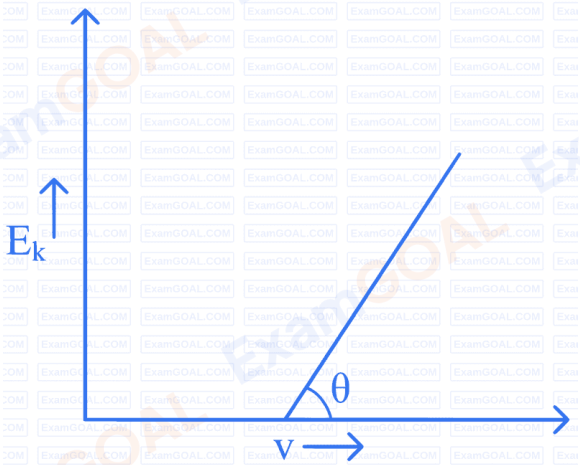
When a metal surface is illuminated by light of wavelength λ, the stopping potential is $8\ \mathrm{V}$. When the same surface is illuminated by light of wavelength 3λ, stopping potential is $2\ \mathrm{V}$. The threshold wavelength for this surface is:

(A) 3λ (B) 9λ (C) 5λ (D) 4.5λ

Open solution

Q48. JEE Main 2024 (Online) 30th January Evening Shift

For the photoelectric effect, the maximum kinetic energy (E_k) of the photoelectrons is plotted against the frequency (v) of the incident photons as shown in figure. The slope of the graph gives



(A) Planck's constant

(B) Work function of the metal

(C) Charge of electron

(D) Ratio of Planck's constant to electric charge

Open solution

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Q49. JEE Main 2024 (Online) 30th January Evening Shift

If the total energy transferred to a surface in time t is 6.48×10^5 J, then the magnitude of the total momentum delivered to this surface for complete absorption will be:

(A) $2.16 \times 10^{-3}\ \mathrm{kg\,m/s}$ (B) $2.46 \times 10^{-3}\ \mathrm{kg\,m/s}$

(C) $1.58 \times 10^{-3}\ \mathrm{kg\,m/s}$ (D) $4.32 \times 10^{-3}\ \mathrm{kg\,m/s}$

Open solution

Q50. JEE Main 2024 (Online) 30th January Morning Shift

The work function of a substance is $3.0\ \mathrm{eV}$. The longest wavelength of light that can cause the emission of photoelectrons from this substance is approximately;

(A) 215 nm (B) 400 nm (C) 414 nm (D) 200 nm

Open solution

Q51. JEE Main 2024 (Online) 29th January Evening Shift

Two sources of light emit with a power of $200\ \mathrm{W}$. The ratio of number of photons of visible light emitted by each source having wavelengths $300\ \mathrm{nm}$ and $500\ \mathrm{nm}$ respectively, will be :

(A) $5:3$ (B) $3:5$ (C) $1:5$ (D) $1:3$

Open solution

Q52. JEE Main 2024 (Online) 29th January Morning Shift

The de-Broglie wavelength of an electron is the same as that of a photon. If velocity of electron is 25% of the velocity of light, then the ratio of K.E. of electron and K.E. of photon will be:

(A) $\frac{1}{4}$ (B) $\frac{8}{1}$ (C) $\frac{1}{8}$ (D) $\frac{1}{1}$

Open solution

Q53. JEE Main 2024 (Online) 27th January Evening Shift

The threshold frequency of a metal with work function $6.63\ \mathrm{eV}$ is :

(A) $16 \times 10^{15}\ \mathrm{Hz}$ (B) $16 \times 10^{12}\ \mathrm{Hz}$ (C) $1.6 \times 10^{15}\ \mathrm{Hz}$ (D) $1.6 \times 10^{12}\ \mathrm{Hz}$

Open solution

Q54. JEE Main 2023 (Online) 15th April Morning Shift

The de Broglie wavelength of an electron having kinetic energy E is λ. If the kinetic energy of electron becomes $\frac{E}{4}$, then its de-Broglie wavelength will be :

(A) $\sqrt{2}\lambda$ (B) 2λ (C) $\frac{\lambda}{2}$ (D) $\frac{\lambda}{\sqrt{2}}$

Open solution

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Q55. JEE Main 2023 (Online) 13th April Evening Shift

Open solution

Given below are two statements:

Statement I : Out of microwaves, infrared rays and ultraviolet rays, ultraviolet rays are the most effective for the emission of electrons from a metallic surface.

Statement II : Above the threshold frequency, the maximum kinetic energy of photoelectrons is inversely proportional to the frequency of the incident light.

In the light of above statements, choose the correct answer form the options given below

(A) Both Statement I and Statement II are true

(B) Statement I is true but statement II is false

(C) Statement I is false but statement II is true

(D) Both Statement I and Statement II are false

Q56. JEE Main 2023 (Online) 13th April Morning Shift

Open solution

The difference between threshold wavelengths for two metal surfaces A and B having work function $\phi_A = 9\,\mathrm{eV}$ and $\phi_B = 4.5\,\mathrm{eV}$ in nm is:

$\{$ Given, hc $= 1242\,\mathrm{eVnm}\}$

(A) 264 (B) 138 (C) 540 (D) 276

Q57. JEE Main 2023 (Online) 12th April Morning Shift

Open solution

A proton and an α-particle are accelerated from rest by $2\,\mathrm{V}$ and $4\,\mathrm{V}$ potentials, respectively. The ratio of their de-Broglie wavelength is :

(A) 4 : 1 (B) 2 : 1 (C) 8 : 1 (D) 16 : 1

Q58. JEE Main 2023 (Online) 11th April Evening Shift

Open solution

The ratio of the de-Broglie wavelengths of proton and electron having same Kinetic energy :

(Assume $m_p = m_e \times 1849$)

(A) 1:43 (B) 1:62 (C) 2:43 (D) 1:30

Q59. JEE Main 2023 (Online) 11th April Morning Shift

Open solution

A metallic surface is illuminated with radiation of wavelength λ, the stopping potential is V_0. If the same surface is illuminated with radiation of wavelength 2λ. the stopping potential becomes $\frac{V_o}{4}$. The threshold wavelength for this metallic surface will be

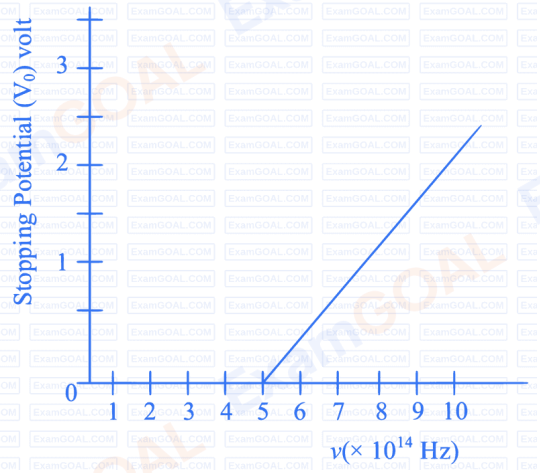
(A) 3λ (B) 4λ (C) $\frac{3}{2}\lambda$ (D) $\frac{\lambda}{4}$

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Q60. JEE Main 2023 (Online) 10th April Evening Shift

Open solution

The variation of stopping potential (V_0) as a function of the frequency (v) of the incident light for a metal is shown in figure. The work function of the surface is



(A) 1.36 eV (B) 18.6 eV (C) 2.98 eV (D) 2.07 eV

Q61. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

The de Broglie wavelength of a molecule in a gas at room temperature (300 K) is λ_1. If the temperature of the gas is increased to 600 K, then the de Broglie wavelength of the same gas molecule becomes

(A) $2\,\lambda_1$ (B) $\frac{1}{2}\lambda_1$

(C) $\frac{1}{\sqrt{2}}\lambda_1$ (D) $\sqrt{2}\,\lambda_1$

Q62. JEE Main 2023 (Online) 8th April Evening Shift

Open solution

In photo electric effect

A. The photocurrent is proportional to the intensity of the incident radiation

B. Maximum Kinetic energy with which photoelectrons are emitted depends on the intensity of incident light.

C. Max. K.E with which photoelectrons are emitted depends on the frequency of incident light.

D. The emission of photoelectrons require a minimum threshold intensity of incident radiation.

E. Max. K.E of the photoelectrons is independent of the frequency of the incident light.

Choose the correct answer from the options given below:

(A) A and E only (B) A and B only (C) B and C only (D) A and C only

Q63. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

Proton (P) and electron (e) will have same de-Broglie wavelength when the ratio of their momentum is (assume, $m_p = 1849\, m_e$):

(A) 1 : 1 (B) 1 : 43 (C) 1 : 1849 (D) 43 : 1

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Q64. JEE Main 2023 (Online) 6th April Evening Shift

The work functions of Aluminium and Gold are $4.1\ \mathrm{eV}$ and $5.1\ \mathrm{eV}$ respectively. The ratio of the slope of the stopping potential versus frequency plot for Gold to that of Aluminium is

(A) 1.5 (B) 1.24 (C) 1 (D) 2

Open solution

Q65. JEE Main 2023 (Online) 6th April Morning Shift

The kinetic energy of an electron, α-particle and a proton are given as $4\ \mathrm{K}, 2\ \mathrm{K}$ and K respectively. The de-Broglie wavelength associated with electron (λe), α-particle $(\lambda\alpha)$ and the proton (λp) are as follows:

(A) $\lambda\alpha < \lambda p < \lambda e$ (B) $\lambda\alpha > \lambda p > \lambda e$

(C) $\lambda\alpha = \lambda p < \lambda e$ (D) $\lambda\alpha = \lambda p > \lambda e$

Open solution

Q66. JEE Main 2023 (Online) 1st February Evening Shift

The threshold frequency of a metal is f_0. When the light of frequency $2f_0$ is incident on the metal plate, the maximum velocity of photoelectrons is v_1. When the frequency of incident radiation is increased to $5\mathrm{f}_0$, the maximum velocity of photoelectrons emitted is v_2. The ratio of v_1 to v_2 is :

(A) $\frac{v_1}{v_2} = \frac{1}{2}$ (B) $\frac{v_1}{v_2} = \frac{1}{16}$ (C) $\frac{v_1}{v_2} = \frac{1}{4}$ (D) $\frac{v_1}{v_2} = \frac{1}{8}$

Open solution

Q67. JEE Main 2023 (Online) 1st February Morning Shift

A proton moving with one tenth of velocity of light has a certain de Broglie wavelength of λ. An alpha particle having certain kinetic energy has the same de-Broglie wavelength λ. The ratio of kinetic energy of proton and that of alpha particle is:

(A) 1 : 4 (B) 2 : 1 (C) 4 : 1 (D) 1 : 2

Open solution

Q68. JEE Main 2023 (Online) 31st January Evening Shift

If the two metals A and B are exposed to radiation of wavelength $350\ \mathrm{nm}$. The work functions of metals A and B are $4.8\mathrm{eV}$ and $2.2\mathrm{eV}$. Then choose the correct option.

(A) Metal B will not emit photo-electrons

(B) Both metals A and B will not emit photo-electrons

(C) Metal A will not emit photo-electrons

(D) Both metals A and B will emit photo-electrons

Open solution

Q69. JEE Main 2023 (Online) 31st January Morning Shift

If a source of electromagnetic radiation having power $15\ \mathrm{kW}$ produces 10^{16} photons per second, the radiation belongs to a part of spectrum is.

(Take Planck constant $h = 6 \times 10^{-34}\mathrm{Js}$)

(A) Gamma rays (B) Radio waves (C) Micro waves (D) Ultraviolet rays

Open solution

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Q70. JEE Main 2023 (Online) 30th January Evening Shift

An electron accelerated through a potential difference V_1 has a de-Broglie wavelength of λ. When the potential is changed to V_2, its de-Broglie wavelength increases by 50%. The value of $\left(\frac{V_1}{V_2}\right)$ is equal to

Open solution

(A) $\frac{3}{2}$ (B) 4 (C) 3 (D) $\frac{9}{4}$

Q71. JEE Main 2023 (Online) 30th January Morning Shift

A small object at rest, absorbs a light pulse of power $20\ \mathrm{mW}$ and duration $300\ \mathrm{ns}$. Assuming speed of light as $3 \times 10^8\ \mathrm{m/s}$, the momentum of the object becomes equal to :

Open solution

(A) $1 \times 10^{-17}\ \mathrm{kg\,m/s}$ (B) $0.5 \times 10^{-17}\ \mathrm{kg\,m/s}$

(C) $3 \times 10^{-17}\ \mathrm{kg\,m/s}$ (D) $2 \times 10^{-17}\ \mathrm{kg\,m/s}$

Q72. JEE Main 2023 (Online) 29th January Evening Shift

The ratio of de-Broglie wavelength of an α particle and a proton accelerated from rest by the same potential is $\frac{1}{\sqrt{m}}$, the value of m is -

Open solution

(A) 2 (B) 16 (C) 8 (D) 4

Q73. JEE Main 2023 (Online) 29th January Morning Shift

The threshold wavelength for photoelectric emission from a material is 5500 $\overset{o}{A}$. Photoelectrons will be emitted, when this material is illuminated with monochromatic radiation from a

Open solution

A. 75 W infra-red lamp

B. 10 W infra-red lamp

C. 75 W ultra-violet lamp

D. 10 W ultra-violet lamp

Choose the correct answer from the options given below :

(A) C only (B) A and D only (C) C and D only (D) B and C only

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Q74. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

Given below are two statements :

Statement I : Stopping potential in photoelectric effect does not depend on the power of the light source.

Statement II : For a given metal, the maximum kinetic energy of the photoelectron depends on the wavelength of the incident light.

In the light of above statements, choose the most appropriate answer from the options given below

(A) Both Statement I and Statement II are incorrect

(B) Statement I is correct but Statement II is incorrect

(C) Both Statement I and Statement II are correct

(D) Statement I is incorrect but Statement II is correct

Q75. JEE Main 2023 (Online) 25th January Morning Shift

Open solution

Electron beam used in an electron microscope, when accelerated by a voltage of 20 kV, has a de-Broglie wavelength of λ_0. IF the voltage is increased to 40 kV, then the de-Broglie wavelength associated with the electron beam would be :

(A) $3\,\lambda_0$ (B) $9\,\lambda_0$ (C) $\frac{\lambda_0}{\sqrt{2}}$ (D) $\frac{\lambda_0}{2}$

Q76. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

An α-particle, a proton and an electron have the same kinetic energy. Which one of the following is correct in case of their de-Broglie wavelength:

(A) $\lambda_\alpha > \lambda_p < \lambda_e$ (B) $\lambda_\alpha > \lambda_p > \lambda_e$

(C) $\lambda_\alpha = \lambda_p = \lambda_e$ (D) $\lambda_\alpha < \lambda_p < \lambda_e$

Q77. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

From the photoelectric effect experiment, following observations are made. Identify which of these are correct.

A. The stopping potential depends only on the work function of the metal.

B. The saturation current increases as the intensity of incident light increases.

C. The maximum kinetic energy of a photo electron depends on the intensity of the incident light.

D. Photoelectric effect can be explained using wave theory of light.

Choose the correct answer from the options given below :

(A) A, B, D only (B) A, C, D only (C) B, C only (D) B only

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Q78. JEE Main 2022 (Online) 30th July Evening Shift

The de Broglie wavelength associated with an electron, accelerated through a potential difference of $15056\ \mathrm{V}$, is:

(A) 0.1 Å (B) 1 Å (C) 10 Å (D) 100 Å

Open solution

Q79. JEE Main 2022 (Online) 29th July Evening Shift

An α particle and a proton are accelerated from rest through the same potential difference. The ratio of linear momenta acquired by above two particles will be:

(A) $\sqrt{2}:1$ (B) $2\sqrt{2}:1$ (C) $4\sqrt{2}:1$ (D) $8:1$

Open solution

Q80. JEE Main 2022 (Online) 29th July Morning Shift

The kinetic energy of emitted electron is E when the light incident on the metal has wavelength λ. To double the kinetic energy, the incident light must have wavelength:

(A) $\frac{hc}{E\lambda-hc}$ (B) $\frac{hc\lambda}{E\lambda+hc}$

(C) $\frac{h\lambda}{E\lambda+hc}$ (D) $\frac{hc\,\lambda}{E\lambda-hc}$

Open solution

Q81. JEE Main 2022 (Online) 28th July Evening Shift

Two streams of photons, possessing energies equal to five and ten times the work function of metal are incident on the metal surface successively. The ratio of maximum velocities of the photoelectron emitted, in the two cases respectively, will be

(A) 1 : 2 (B) 1 : 3 (C) 2 : 3 (D) 3 : 2

Open solution

Q82. JEE Main 2022 (Online) 28th July Morning Shift

The equation $\lambda=\frac{1.227}{x}\ \mathrm{nm}$ can be used to find the de-Brogli wavelength of an electron. In this equation x stands for :

Where

$\mathrm{m}=$ mass of electron

$\mathrm{P}=$ momentum of electron

$\mathrm{K}=$ Kinetic energy of electron

$\mathrm{V}=$ Accelerating potential in volts for electron

(A) $\sqrt{\mathrm{mK}}$ (B) $\sqrt{\mathrm{P}}$ (C) $\sqrt{\mathrm{K}}$ (D) $\sqrt{\mathrm{V}}$

Open solution

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Q83. JEE Main 2022 (Online) 27th July Evening Shift

Open solution

With reference to the observations in photo-electric effect, identify the correct statements from below :

(A) The square of maximum velocity of photoelectrons varies linearly with frequency of incident light.

(B) The value of saturation current increases on moving the source of light away from the metal surface.

(C) The maximum kinetic energy of photo-electrons decreases on decreasing the power of LED (light emitting diode) source of light.

(D) The immediate emission of photo-electrons out of metal surface can not be explained by particle nature of light/electromagnetic waves.

(E) Existence of threshold wavelength can not be explained by wave nature of light/ electromagnetic waves.

Choose the correct answer from the options given below :

(A) (A) and (B) only **(B)** (A) and (E) only **(C)** (C) and (E) only **(D)** (D) and (E) only

Q84. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

An electron (mass m) with an initial velocity $\vec{v} = v_0\hat{i}\,(v_0 > 0)$ is moving in an electric field $\vec{E} = -E_0\hat{i}\,(E_0 > 0)$ where E_0 is constant. If at $\mathrm{t} = 0$ de Broglie wavelength is $\lambda_0 = \frac{h}{mv_0}$, then its de Broglie wavelength after time t is given by

(A) λ_0

(B) $\lambda_0\left(1 + \frac{eE_0t}{mv_0}\right)$

(C) $\lambda_0 t$

(D) $\frac{\lambda_0}{\left(1+\frac{eE_0t}{mv_0}\right)}$

Q85. JEE Main 2022 (Online) 26th July Evening Shift

Open solution

A nucleus of mass M at rest splits into two parts having masses $\frac{M'}{3}$ and $\frac{2M'}{3}(M' < M)$. The ratio of de Broglie wavelength of two parts will be :

(A) 1 : 2 **(B)** 2 : 1 **(C)** 1 : 1 **(D)** 2 : 3

Q86. JEE Main 2022 (Online) 26th July Morning Shift

Open solution

A parallel beam of light of wavelength $900\ \mathrm{nm}$ and intensity $100\ \mathrm{Wm}^{-2}$ is incident on a surface perpendicular to the beam. The number of photons crossing $1\ \mathrm{cm}^2$ area perpendicular to the beam in one second is :

(A) 3×10^{16} **(B)** 4.5×10^{16} **(C)** 4.5×10^{17} **(D)** 4.5×10^{20}

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Q87. JEE Main 2022 (Online) 25th July Evening Shift

The ratio of wavelengths of proton and deuteron accelerated by potential V_p and V_d is $1 : \sqrt{2}$. Then the ratio of V_p to V_d will be :

(A) 1 : 1 (B) $\sqrt{2} : 1$ (C) 2 : 1 (D) 4 : 1

Open solution

Q88. JEE Main 2022 (Online) 25th July Morning Shift

A metal exposed to light of wavelength $800\ \text{nm}$ and emits photoelectrons with a certain kinetic energy. The maximum kinetic energy of photo-electron doubles when light of wavelength $500\ \text{nm}$ is used. The workfunction of the metal is : (Take hc $= 1230\ \text{eV} - \text{nm}$).

(A) 1.537 eV (B) 2.46 eV (C) 0.615 eV (D) 1.23 eV

Open solution

Q89. JEE Main 2022 (Online) 30th June Morning Shift

A source of monochromatic light liberates 9×10^{20} photon per second with wavelength 600 nm when operated at 400 W. The number of photons emitted per second with wavelength of 800 nm by the source of monochromatic light operating at same power will be :

(A) 12×10^{20} (B) 6×10^{20} (C) 9×10^{20} (D) 24×10^{20}

Open solution

Q90. JEE Main 2022 (Online) 29th June Evening Shift

The electric field at a point associated with a light wave is given by

E = 200 [sin (6×10^{15})t + sin (9×10^{15})t] Vm^{-1}

Given : h = 4.14×10^{-15} eVs

If this light falls on a metal surface having a work function of 2.50 eV, the maximum kinetic energy of the photoelectrons will be

(A) 1.90 eV (B) 3.27 eV (C) 3.60 eV (D) 3.42 eV

Open solution

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Q91. JEE Main 2022 (Online) 29th June Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : The photoelectric effect does not takes place, if the energy of the incident radiation is less than the work function of a metal.

Reason R : Kinetic energy of the photoelectrons is zero, if the energy of the incident radiation is equal to the work function of a metal.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Both A and R are correct and R is the correct explanation of A.

(B) Both A and R are correct but R is not the correct explanation of A.

(C) A is correct but R is not correct.

(D) A is not correct but R is correct.

Q92. JEE Main 2022 (Online) 28th June Evening Shift

Open solution

Let K_1 and K_2 be the maximum kinetic energies of photo-electrons emitted when two monochromatic beams of wavelength λ_1 and λ_2, respectively are incident on a metallic surface. If $\lambda_1 = 3\lambda_2$ then :

(A) $K_1 > \frac{K_2}{3}$ (B) $K_1 < \frac{K_2}{3}$ (C) $K_1 = \frac{K_2}{3}$ (D) $K_2 = \frac{K_1}{3}$

Q93. JEE Main 2022 (Online) 28th June Morning Shift

Open solution

The de Broglie wavelengths for an electron and a photon are λ_e and λ_p respectively. For the same kinetic energy of electron and photon, which of the following presents the correct relation between the de Broglie wavelengths of two ?

(A) $\lambda_p \propto \lambda_e^2$ (B) $\lambda_p \propto \lambda_e$

(C) $\lambda_p \propto \sqrt{\lambda_e}$ (D) $\lambda_p \propto \sqrt{\frac{1}{\lambda_e}}$

Q94. JEE Main 2022 (Online) 27th June Morning Shift

Open solution

An α particle and a carbon 12 atom has same kinetic energy K. The ratio of their de-Broglie wavelengths $(\lambda_\alpha : \lambda_{C12})$ is :

(A) $1 : \sqrt{3}$ (B) $\sqrt{3} : 1$ (C) $3 : 1$ (D) $2 : \sqrt{3}$

Q95. JEE Main 2022 (Online) 26th June Evening Shift

Open solution

A metal surface is illuminated by a radiation of wavelength 4500 $\overset{o}{A}$. The ejected photo-electron enters a constant magnetic field of 2 mT making an angle of 90° with the magnetic field. If it starts revolving in a circular path of radius 2 mm, the work function of the metal is approximately :

(A) 1.36 eV (B) 1.69 eV (C) 2.78 eV (D) 2.23 eV

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Q96. JEE Main 2022 (Online) 26th June Morning Shift

An electron with speed v and a photon with speed c have the same de-Broglie wavelength. If the kinetic energy and momentum of electron are E_e and p_e and that of photon are E_{ph} and p_{ph} respectively. Which of the following is correct?

Open solution

(A) $\frac{E_e}{E_{ph}} = \frac{2c}{v}$

(B) $\frac{E_e}{E_{ph}} = \frac{v}{2c}$

(C) $\frac{p_e}{p_{ph}} = \frac{2c}{v}$

(D) $\frac{p_e}{p_{ph}} = \frac{v}{2c}$

Q97. JEE Main 2022 (Online) 25th June Evening Shift

A proton, a neutron, an electron and an α particle have same energy. If λ_p, λ_n, λ_e and λ_α are the de Broglie's wavelengths of proton, neutron, electron and α particle respectively, then choose the correct relation from the following :

Open solution

(A) $\lambda_p = \lambda_n > \lambda_e > \lambda_\alpha$

(B) $\lambda_\alpha < \lambda_n < \lambda_p < \lambda_e$

(C) $\lambda_e < \lambda_p = \lambda_n > \lambda_\alpha$

(D) $\lambda_e = \lambda_p = \lambda_n = \lambda_\alpha$

Q98. JEE Main 2022 (Online) 24th June Evening Shift

The light of two different frequencies whose photons have energies 3.8 eV and 1.4 eV respectively, illuminate a metallic surface whose work function is 0.6 eV successively. The ratio of maximum speeds of emitted electrons for the two frequencies respectively will be :

Open solution

(A) 1 : 1

(B) 2 : 1

(C) 4 : 1

(D) 1 : 4

Q99. JEE Main 2021 (Online) 1st September Evening Shift

The temperature of an ideal gas in 3-dimensions is 300 K. The corresponding de-Broglie wavelength of the electron approximately at 300 K, is :

Open solution

[m_e = mass of electron = 9×10^{-31} kg, h = Planck constant = 6.6×10^{-34} Js, k_B = Boltzmann constant = 1.38×10^{-23} JK^{-1}]

(A) 6.26 nm

(B) 8.46 nm

(C) 2.26 nm

(D) 3.25 nm

Q100. JEE Main 2021 (Online) 31st August Evening Shift

A free electron of 2.6 eV energy collides with a H^+ ion. This results in the formation of a hydrogen atom in the first excited state and a photon is released. Find the frequency of the emitted photon. (h = 6.6×10^{-34} Js)

Open solution

(A) 1.45×10^{16} MHz

(B) 0.19×10^{15} MHz

(C) 1.45×10^{9} MHz

(D) 9.0×10^{27} MHz

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Q101. JEE Main 2021 (Online) 31st August Evening Shift

Consider two separate ideal gases of electrons and protons having same number of particles. The temperature of both the gases is the same. The ratio of the uncertainty in determining the position of an electron to that of a proton is proportional to :-

(A) $\left(\frac{m_p}{m_e}\right)^{3/2}$

(B) $\sqrt{\frac{m_e}{m_p}}$

(C) $\sqrt{\frac{m_p}{m_e}}$

(D) $\frac{m_p}{m_e}$

Open solution

Q102. JEE Main 2021 (Online) 31st August Morning Shift

A moving proton and electron have the same de-Broglie wavelength. If K and P denote the K.E. and momentum respectively. Then choose the correct option :

(A) $K_p < K_e$ and $P_p = P_e$

(B) $K_p = K_e$ and $P_p = P_e$

(C) $K_p < K_e$ an $P_p < P_e$

(D) $K_p > K_e$ and $P_p = P_e$

Open solution

Q103. JEE Main 2021 (Online) 27th August Evening Shift

A monochromatic neon lamp with wavelength of 670.5 nm illuminates a photo-sensitive material which has a stopping voltage of 0.48 V. What will be the stopping voltage if the source light is changed with another source of wavelength of 474.6 nm?

(A) 0.96 V

(B) 1.25 V

(C) 0.24 V

(D) 1.5 V

Open solution

Q104. JEE Main 2021 (Online) 27th August Morning Shift

In a photoelectric experiment, increasing the intensity of incident light :

(A) increases the number of photons incident and also increases the K.E. of the ejected electrons

(B) increases the frequency of photons incident and increases the K.E. of the ejected electrons

(C) increases the frequency of photons incident and the K.E. of the ejected electrons remains unchanged

(D) increases the number of photons incident and the K.E. of the ejected electrons remains unchanged

Open solution

Q105. JEE Main 2021 (Online) 26th August Evening Shift

The de-Broglie wavelength of a particle having kinetic energy E is λ. How much extra energy must be given to this particle so that the de-Broglie wavelength reduces to 75% of the initial value?

(A) $\frac{1}{9}$E

(B) $\frac{7}{9}$E

(C) E

(D) $\frac{16}{9}$E

Open solution

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Q106. JEE Main 2021 (Online) 26th August Morning Shift

In a photoelectric experiment ultraviolet light of wavelength 280 nm is used with lithium cathode having work function ϕ = 2.5 eV. If the wavelength of incident light is switched to 400 nm, find out the change in the stopping potential. (h = 6.63 $\times$ 10^{-34} Js, c = 3 $\times$ 10^{8} ms^{-1})

Open solution

(A) 1.3 V (B) 1.1 V (C) 1.9 V (D) 0.6 V

Q107. JEE Main 2021 (Online) 27th July Evening Shift

An electron and proton are separated by a large distance. The electron starts approaching the proton with energy 3 eV. The proton captures the electron and forms a hydrogen atom in second excited state. The resulting photon is incident on a photosensitive metal of threshold wavelength 4000$\overset{o}{A}$. What is the maximum kinetic energy of the emitted photoelectron?

Open solution

(A) 7.61 eV (B) 1.41 eV

(C) 3.3 eV (D) No photoelectron would be emitted

Q108. JEE Main 2021 (Online) 25th July Evening Shift

An electron moving with speed v and a photon moving with speed c, have same de-Broglie wavelength. The ratio of kinetic energy of electron to that of photon is :

Open solution

(A) $\frac{3c}{v}$ (B) $\frac{v}{3c}$ (C) $\frac{v}{2c}$ (D) $\frac{2c}{v}$

Q109. JEE Main 2021 (Online) 25th July Evening Shift

When radiation of wavelength λ is incident on a metallic surface, the stopping potential of ejected photoelectrons is 4.8 V. If the same surface is illuminated by radiation of double the previous wavelength, then the stopping potential becomes 1.6 V. The threshold wavelength of the metal is :

Open solution

(A) 2λ (B) 4λ (C) 8λ (D) 6λ

Q110. JEE Main 2021 (Online) 25th July Morning Shift

What should be the order of arrangement of de-Broglie wavelength of electron (λ_e), an α-particle (λ_a) and proton (λ_p) given that all have the same kinetic energy?

Open solution

(A) $\lambda_e = \lambda_p = \lambda_\alpha$ (B) $\lambda_e < \lambda_p < \lambda_\alpha$

(C) $\lambda_e > \lambda_p > \lambda_\alpha$ (D) $\lambda_e = \lambda_p > \lambda_\alpha$

Q111. JEE Main 2021 (Online) 25th July Morning Shift

A particle of mass 4M at rest disintegrates into two particles of mass M and 3M respectively having non zero velocities. The ratio of de-Broglie wavelength of particle of mass M to that of mass 3M will be :

Open solution

(A) 1 : 3 (B) 3 : 1 (C) 1 : $\sqrt{3}$ (D) 1 : 1

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Q112. JEE Main 2021 (Online) 22th July Evening Shift

An electron of mass m_e and a proton of mass m_p are accelerated through the same potential difference. The ratio of the de-Broglie wavelength associated with the electron to that with the proton is

(A) $\frac{m_e}{m_p}$ (B) 1 (C) $\frac{m_p}{m_e}$ (D) $\sqrt{\frac{m_p}{m_e}}$

Open solution

Q113. JEE Main 2021 (Online) 20th July Evening Shift

An electron having de-Broglie wavelength λ is incident on a target in a X-ray tube. Cut-off wavelength of emitted X-ray is :

(A) 0 (B) $\frac{2m^2c^2\lambda^2}{h^2}$

(C) $\frac{2mc\lambda^2}{h}$ (D) $\frac{hc}{mc}$

Open solution

Q114. JEE Main 2021 (Online) 20th July Morning Shift

The radiation corresponding to $3 \rightarrow 2$ transition of a hydrogen atom falls on a gold surface to generate photoelectrons. The electrons are passed through a magnetic field of 5×10^{-4} T. Assume that the radius of the largest circular path followed by these electrons is 7 mm, the work function of the metal is : (Mass of electron = 9.1×10^{-31} kg)

(A) 1.36 eV (B) 1.88 eV (C) 0.82 eV (D) 0.16 eV

Open solution

Q115. JEE Main 2021 (Online) 18th March Morning Shift

A particle is travelling 4 time as fast as an electron. Assuming the ratio of de-Broglie wavelength of a particle to that of electron is 2 : 1, the mass of the particle is :

(A) $\frac{1}{16}$ times the mass of e^- (B) 8 times the mass of e^-

(C) 16 times the mass of e^- (D) $\frac{1}{8}$ times the mass of e^-

Open solution

Q116. JEE Main 2021 (Online) 17th March Evening Shift

Two identical photocathodes receive the light of frequencies f_1 and f_2 respectively. If the velocities of the photo-electrons coming out are v_1 and v_2 respectively, then

(A) $v_1 - v_2 = \left[\frac{2h}{m}(f_1 - f_2)\right]^{\frac{1}{2}}$ (B) $v_1^2 + v_2^2 = \frac{2h}{m}[f_1 + f_2]$

(C) $v_1 + v_2 = \left[\frac{2h}{m}(f_1 + f_2)\right]^{\frac{1}{2}}$ (D) $v_1^2 - v_2^2 = \frac{2h}{m}[f_1 - f_2]$

Open solution

Q117. JEE Main 2021 (Online) 17th March Morning Shift

An electron of mass m and a photon have same energy E. The ratio of wavelength of electron to that of photon is : (c being the velocity of light)

(A) $\frac{1}{c}\left(\frac{2m}{E}\right)^{1/2}$ (B) $\frac{1}{c}\left(\frac{E}{2m}\right)^{1/2}$

(C) $\left(\frac{E}{2m}\right)^{1/2}$ (D) $c(2mE)^{1/2}$

Open solution

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Q118. JEE Main 2021 (Online) 16th March Evening Shift

The de-Broglie wavelength associated with an electron and a proton were calculated by accelerating them through same potential of 100 V. What should nearly be the ratio of their wavelengths? (m_p = 1.00727u m_e = 0.00055u)

Open solution

(A) 41.4 : 1 (B) $(1860)^2 : 1$ (C) 1860 : 1 (D) 43 : 1

Q119. JEE Main 2021 (Online) 16th March Morning Shift

The stopping potential in the context of photoelectric effect depends on the following property of incident electromagnetic radiation :

Open solution

(A) Phase (B) Frequency (C) Amplitude (D) Intensity

Q120. JEE Main 2021 (Online) 26th February Evening Shift

The recoil speed of a hydrogen atom after it emits a photon in going from n = 5 state to n = 1 state will be :

Open solution

(A) 4.34 m/s (B) 2.19 m/s (C) 3.25 m/s (D) 4.17 m/s

Q121. JEE Main 2021 (Online) 26th February Morning Shift

Given below are two statements : one is labeled as Assertion A and the other is labelled as Reason R.

Open solution

Assertion A : An electron microscope can achieve better resolving power than an optical microscope.

Reason R : The de Broglie's wavelength of the electrons emitted from an electron gun is much less than wavelength of visible light.

In the light of the above statements, choose the correct answer from the options given below :

(A) A is false but R is true.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) Both A and R are true and R is the correct explanation of A.

(D) A is true but R is false.

Q122. JEE Main 2021 (Online) 25th February Evening Shift

An electron of mass m_e and a proton of mass m_p = 1836 m_e are moving with the same speed. The ratio of their de Broglie wavelength $\frac{\lambda_{electron}}{\lambda_{proton}}$ will be :

Open solution

(A) 1 (B) 1836 (C) $\frac{1}{1836}$ (D) 918

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Q123. JEE Main 2021 (Online) 25th February Evening Shift

The stopping potential for electrons emitted from a photosensitive surface illuminated by light of wavelength 491 nm is 0.710 V. When the incident wavelength is changed to a new value, the stopping potential is 1.43 V. The new wavelength is :

Open solution

(A) 400 nm (B) 329 nm (C) 309 nm (D) 382 nm

Q124. JEE Main 2021 (Online) 25th February Morning Shift

An α particle and a proton are accelerated from rest by a potential difference of 200V. After this, their de Broglie wavelengths are λ_α and λ_p respectively. The ratio $\frac{\lambda_p}{\lambda_\alpha}$ is :

Open solution

(A) 8 (B) 2.8 (C) 7.8 (D) 3.8

Q125. JEE Main 2021 (Online) 24th February Evening Shift

An X-ray tube is operated at 1.24 million volt. The shortest wavelength of the produced photon will be :

Open solution

(A) 10^{-2} nm (B) 10^{-1} nm (C) 10^{-3} nm (D) 10^{-4} nm

Q126. JEE Main 2021 (Online) 24th February Evening Shift

The de-Broglie wavelength of a proton and α-particle are equal. The ratio of their velocities is :

Open solution

(A) 4 : 2 (B) 4 : 3 (C) 4 : 1 (D) 1 : 4

Q127. JEE Main 2021 (Online) 24th February Morning Shift

Given below are two statements :

Open solution

Statement I : Two photons having equal linear momenta have equal wavelengths.

Statement II : If the wavelength of photon is decreased, then the momentum and energy of a photon will also decrease.

In the light of the above statements, choose the correct answer from the options given below.

(A) Statement I is false but Statement II is true
(B) Both Statement I and Statement II are false
(C) Both Statement I and Statement II are true
(D) Statement I is true but Statement II is false

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Q128. JEE Main 2020 (Online) 6th September Evening Slot

Open solution

Assuming the nitrogen molecule is moving with r.m.s. velocity at 400 K, the de-Broglie wavelength of nitrogen molecule is close to :

(Given : nitrogen molecule weight : 4.64×10^{-26} kg,

Boltzmann constant: 1.38×10^{-23} J/K,

Planck constant : 6.63×10^{-34} J.s)

(A) 0.44 $\overset{o}{A}$ (B) 0.34 $\overset{o}{A}$ (C) 0.20 $\overset{o}{A}$ (D) 0.24 $\overset{o}{A}$

Q129. JEE Main 2020 (Online) 6th September Morning Slot

Open solution

An electron, a doubly ionized helium ion (He^{++}) and a proton are having the same kinetic energy. The relation between their respective de-Broglie wavelengths λ_e, $\lambda_{He^{++}}$ and λ_p is :

(A) $\lambda_e > \lambda_{He^{++}} > \lambda_p$

(B) $\lambda_e < \lambda_p < \lambda_{He^{++}}$

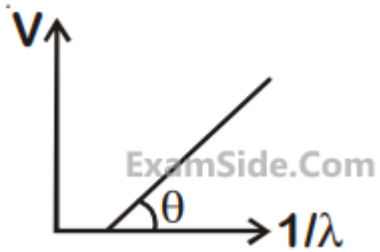
(C) $\lambda_e > \lambda_p > \lambda_{He^{++}}$

(D) $\lambda_e < \lambda_{He^{++}} = \lambda_p$

Q130. JEE Main 2020 (Online) 4th September Evening Slot

Open solution

In a photoelectric effect experiment, the graph of stopping potential V versus reciprocal of wavelength obtained is shown in the figure. As the intensity of incident radiation is increased :



(A) Slope of the straight line get more steep

(B) Graph does not change

(C) Straight line shifts to left

(D) Straight line shifts to right

Q131. JEE Main 2020 (Online) 4th September Morning Slot

Open solution

Particle A of mass $m_A = \frac{m}{2}$ moving along the x-axis with velocity v_0 collides elastically with another particle B at rest having mass $m_B = \frac{m}{3}$. If both particles move along the x-axis after the collision, the change $\Delta\lambda$ in de-Broglie wavlength of particle A, in terms of its de-Broglie wavelength (λ_0) before collision is :

(A) $\Delta\lambda = \frac{5}{2}\lambda_0$

(B) $\Delta\lambda = \frac{3}{2}\lambda_0$

(C) $\Delta\lambda = 2\lambda_0$

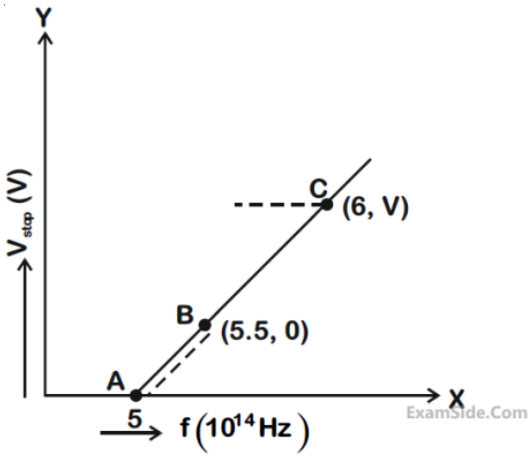
(D) $\Delta\lambda = 4\lambda_0$

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Q132. JEE Main 2020 (Online) 4th September Morning Slot

Open solution

Given figure shows few data points in a photoelectric effect experiment for a certain metal. The minimum energy for ejection of electron from its surface is:
(Planck's constant h = 6.62×10^{-34} J.s)



(A) 2.10 eV (B) 2.27 eV (C) 2.59 eV (D) 1.93 eV

Q133. JEE Main 2020 (Online) 3rd September Evening Slot

Open solution

Two sources of light emit X-rays of wavelength 1 nm and visible light of wavelength 500 nm, respectively. Both the sources emit light of the same power 200 W. The ratio of the number density of photons of X-rays to the number density of photons of the visible light of the given wavelengths is :

(A) $\frac{1}{500}$ (B) 500 (C) 250 (D) $\frac{1}{250}$

Q134. JEE Main 2020 (Online) 3rd September Morning Slot

Open solution

When the wavelength of radiation falling on a metal is changed from 500 nm to 200 nm, the maximum kinetic energy of the photoelectrons becomes three times larger. The work function of the metal is close to :

(A) 1.02 eV (B) 0.81 eV (C) 0.61 eV (D) 0.52 eV

Q135. JEE Main 2020 (Online) 2nd September Evening Slot

Open solution

A particle is moving 5 times as fast as an electron. The ratio of the de-Broglie wavelength of the particle to that of the electron is 1.878×10^{-4}. The mass of the particle is close to

(A) 1.2×10^{-28} kg (B) 9.1×10^{-31} kg
(C) 4.8×10^{-27} kg (D) 9.7×10^{-28} kg

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Q136. JEE Main 2020 (Online) 9th January Evening Slot

An electron of mass m and magnitude of charge |e| initially at rest gets accelerated by a constant electric field E. The rate of change of de-Broglie wavelength of this electron at time t ignoring relativistic effects is :

Open solution

(A) $\frac{-h}{|e|Et}$

(B) $\frac{-h}{|e|E\sqrt{t}}$

(C) $\frac{-h}{|e|Et^2}$

(D) $\frac{|e|Et}{h}$

Q137. JEE Main 2020 (Online) 9th January Morning Slot

A particle moving with kinetic energy E has de Broglie wavelength λ. If energy ΔE is added to its energy, the wavelength become $\lambda/2$. Value of ΔE, is :

Open solution

(A) E

(B) 3E

(C) 2E

(D) 4E

Q138. JEE Main 2020 (Online) 9th January Morning Slot

Radiation, with wavelength 6561 $\overset{o}{A}$ falls on a metal surface to produce photoelectrons. The electrons are made to enter a uniform magnetic field of 3×10^{-4} T. If the radius of the largest circular path followed by the electrons is 10 mm, the work function of the metal is close to :

Open solution

(A) 1.8eV

(B) 0.8eV

(C) 1.1eV

(D) 1.6eV

Q139. JEE Main 2020 (Online) 8th January Evening Slot

An electron (mass m) with initial velocity $\vec{v} = v_0\hat{i} + v_0\hat{j}$ is in an electric field $\vec{E} = -E_0\hat{k}$. If λ_0 is initial de-Broglie wavelength of electron, its de-Broglie wave length at time t is given by :

Open solution

(A) $\frac{\lambda_0}{\sqrt{1+\frac{e^2E^2t^2}{m^2v_0^2}}}$

(B) $\frac{\lambda_0\sqrt{2}}{\sqrt{1+\frac{e^2E^2t^2}{m^2v_0^2}}}$

(C) $\frac{\lambda_0}{\sqrt{1+\frac{e^2E^2t^2}{2m^2v_0^2}}}$

(D) $\frac{\lambda_0}{\sqrt{2+\frac{e^2E^2t^2}{m^2v_0^2}}}$

Q140. JEE Main 2020 (Online) 8th January Morning Slot

When photon of energy 4.0 eV strikes the surface of a metal A, the ejected photoelectrons have maximum kinetic energy T_A eV and de-Broglie wavelength λ_A. The maximum kinetic energy of photoelectrons liberated from another metal B by photon of energy 4.50 eV is $T_B = (T_A - 1.5)$ eV. If the de-Broglie wavelength of these photoelectrons $\lambda_B = 2\lambda_A$, then the work function of metal B is :

Open solution

(A) 1.5eV

(B) 4eV

(C) 2eV

(D) 3eV

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Q141. JEE Main 2020 (Online) 7th January Evening Slot

An electron (of mass m) and a photon have the same energy E in the range of a few eV. The ratio of the de-Broglie wavelength associated with the electron and the wavelength of the photon is (c = speed of light in vaccuum)

Open solution

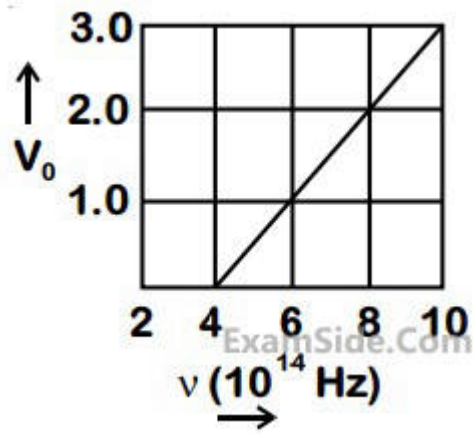
(A) $\frac{1}{c}\left(\frac{2E}{m}\right)^{\frac{1}{2}}$ (B) $\frac{1}{c}\left(\frac{E}{2m}\right)^{\frac{1}{2}}$

(C) $\left(\frac{E}{2m}\right)^{\frac{1}{2}}$ (D) $c(2mE)^{\frac{1}{2}}$

Q142. JEE Main 2019 (Online) 12th April Morning Slot

The stopping potential V_0 (in volt) as a function of frequency (υ) for a sodium emitter, is shown in the figure. The work function of sodium, from the data plotted in the figure, will be:
(Given: Planck's constant (h) = 6.63 × 10^{-34} Js, electron charges e = 1.6 × 10^{-19} C)

Open solution



(A) 1.95 eV (B) 2.12 eV (C) 1.82 eV (D) 1.66 eV

Q143. JEE Main 2019 (Online) 10th April Evening Slot

A 2 mW laser operates at wavelength of 500 nm. The number of photons that will be emitted per second is : [Given Planck's constant h = 6.6 × 10^{-34} Js, speed of light c = 3.0 × 10^8 m/s]

Open solution

(A) 5 × 10^{15} (B) 1.5 × 10^{16} (C) 1 × 10^{16} (D) 2 × 10^{16}

Q144. JEE Main 2019 (Online) 10th April Morning Slot

In a photoelectric effect experiment the threshold wavelength of the light is 380 nm. If the wavelentgh of incident light is 260 nm, the maximum kinetic energy of emitted electrons will be:
Given E (in eV) = 1237/λ (in nm)

Open solution

(A) 4.5 eV (B) 15.1 eV (C) 3.0 eV (D) 1.5 eV

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Q145. JEE Main 2019 (Online) 9th April Evening Slot

A particle 'P' is formed due to a completely inelastic collision of particles 'x' and 'y' having de-Broglie wavelengths 'λ_x' and 'λ_y' respectively. If x and y were moving in opposite directions, then the de-Broglie wavelength of 'P' is :-

Open solution

(A) $\lambda_x - \lambda_y$

(B) $\frac{\lambda_x \lambda_y}{|\lambda_x - \lambda_y|}$

(C) $\lambda_x + \lambda_y$

(D) $\frac{\lambda_x \lambda_y}{\lambda_x + \lambda_y}$

Q146. JEE Main 2019 (Online) 9th April Morning Slot

The electric field of light wave is given as

$$\overrightarrow{E} = 10^{-3} \cos\left(\frac{2\pi x}{5 \times 10^{-7}} - 2\pi \times 6 \times 10^{14} t\right) \hat{x} \frac{\text{N}}{C}$$

Open solution

This light falls on a metal plate of work function 2eV. The stopping potential of the photoelectrons is :

Given, E (in eV) = 12375/λ(inÅ)

(A) 2.48 V (B) 0.48 V (C) 0.72 V (D) 2.0 V

Q147. JEE Main 2019 (Online) 8th April Evening Slot

A nucleus A, with a finite de-broglie wavelength λ_A, undergoes spontaneous fission into two nuclei B and C of equal mass. B flies in the same direction as that of A, while C flies in the opposite direction with a velocity equal to half of that of B. The de-Broglie wavelengths λ_B and λ_C of B and C are respectively :

Open solution

(A) λ_A, $2\lambda_A$ (B) $2\lambda_A$, λ_A (C) λ_A, $\lambda_A/2$ (D) $\lambda_A/2$, λ_A

Q148. JEE Main 2019 (Online) 8th April Morning Slot

Two particles move at right angle to each other. Their de-Broglie wavelengths are λ_1 and λ_2 respectively. The particles suffer perfectly inelastic collision. The de-Broglie wavelength λ of the final particle, is given by :

Open solution

(A) $\lambda = \frac{\lambda_1 + \lambda_2}{2}$ (B) $\frac{1}{\lambda^2} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2}$

(C) $\lambda = \sqrt{\lambda_1 \lambda_2}$ (D) $\frac{2}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

Q149. JEE Main 2019 (Online) 12th January Evening Slot

In a Frank-Hertz experiment, an electron of energy 5.6 eV passes through mercury vapour and emerges with an energy 0.7 eV. The minimum wavelength of photons emitted by mercury atoms is close to :

Open solution

(A) 2020 nm (B) 250 nm (C) 1700 nm (D) 220 nm

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Q150. JEE Main 2019 (Online) 12th January Evening Slot

When a certain photosensitive surface is illuminated with monochromatic light of frequency v, the stopping potential for the current is $-V_0/2$. When the surface is illuminated by monochromatic light of frequency v/2, the stopping potential is $-V_0$. The threshold frequency for photoelectric emission is :

Open solution

(A) $2v$ (B) $\frac{4}{3}v$ (C) $\frac{3v}{2}$ (D) $\frac{5v}{3}$

Q151. JEE Main 2019 (Online) 12th January Morning Slot

A particle A of mass 'm' and charge 'q' is accelerated by a potential difference of 50 V. Another particle B of mass ' 4 m' and charge 'q' is accelerated by a potential difference of 2500 V. The ratio of de-Broglie wavelengths $\frac{\lambda_A}{\lambda_B}$ is close to :

Open solution

(A) 4.47 (B) 10.00 (C) 14.14 (D) 0.07

Q152. JEE Main 2019 (Online) 11th January Evening Slot

In a photoelectric experiment, the wavelength of the light incident on a metal is changed from 300 nm to 400 nm. The decrease in the stopping potential is close to: ($\frac{hc}{e}$ = 1240 nm eV)

Open solution

(A) 0.5 V (B) 1.0 V (C) 2.0 V (D) 1.5 V

Q153. JEE Main 2019 (Online) 11th January Morning Slot

If the de Broglie wavelength of an electron is equal to 10^{-3} times the wavelength of a photon of frequency 6×10^{14} Hz, then the speed of electron is equal to : (Speed of light = 3×10^8 m/s, Planck's constant = 6.63×10^{-34} J.s, Mass of electron = 9.1×10^{-31} kg)

Open solution

(A) 1.7×10^6 m/s (B) 1.45×10^6 m/s

(C) 1.1×10^6 m/s (D) 1.8×10^6 m/s

Q154. JEE Main 2019 (Online) 10th January Evening Slot

A metal plate of area 1×10^{-4} m^2 is illuminated by a radiation of intensity 16 mW/m^2. The work function of the metal is 5 eV. The energy of the incident photons is 10 eV and only 10% of it produces photo electrons. The number of emitted photoelectrons per second and their maximum energy, respectively, will be

Open solution

(A) 10^{14} and 10 eV (B) 10^{12} and 5 eV (C) 10^{11} and 5 eV (D) 10^{10} and 5 eV

Q155. JEE Main 2019 (Online) 10th January Morning Slot

In an electron microscope, the resolution that can be achieved is of the order of the wavelength of electrons used. To resolve a width of 7.5×10^{-12} m, the minimum electron energy required is close to -

Open solution

(A) 25 keV (B) 500 keV (C) 100 keV (D) 1 keV

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Q156. JEE Main 2019 (Online) 9th January Evening Slot

The magnetic field associated with a light wave is given, at the origin, by B = B_0 [sin(3.14 $\times$ 10^7)ct + sin(6.28 $\times$ 10^7)ct]. If this light falls on a silver plate having a work function of 4.7 eV, what will be the maximum kinetic energy of the photo electrons ?
(Take c = 3 $\times$ 10^8 ms^{-1}, h = 6.6 $\times$ 10^{-34}J-s)

(A) 6.82 eV (B) 12.5 eV (C) 8.52 eV (D) 7.72 eV

Open solution

Q157. JEE Main 2019 (Online) 9th January Morning Slot

Surface of certain metal is first illuminated with light of wavelength λ_1 = 350 nm and then, by light of wavelength λ_2 = 540 nm. It is found that the maximum speed of the photo electrons in the two cases differ by a factor of 2. The work function of the metal (in eV) is close to :

(Energy of photon = $\frac{1240}{\lambda(in\ nm)}$eV)

(A) 1.8 (B) 2.5 (C) 5.6 (D) 1.4

Open solution

Q158. JEE Main 2018 (Online) 16th April Morning Slot

The de-Broglie wavelength (λ_B) associated with the electron orbiting in the second excited state of hydrogen atom is related to that in the ground state (λ_G) by :

(A) $\lambda_B = 2\lambda_G$ (B) $\lambda_B = 3\lambda_G$ (C) $\lambda_B = \lambda_{G/2}$ (D) $\lambda_B = \lambda_{G/3}$

Open solution

Q159. JEE Main 2018 (Online) 15th April Evening Slot

If the de Broglie wavelengths associated with a proton and an α-particle are equal, then the ratio of velocities of the proton and the α-particle will be :

(A) 4 : 1 (B) 2 : 1 (C) 1 : 2 (D) 1 : 4

Open solution

Q160. JEE Main 2018 (Online) 15th April Morning Slot

Two electrons are moving with non-relativistic speed perpendicular to each other. If corresponding de Broglie wavelength are λ_1 and λ_2, their de Broglie wavelength in the frame of reference attached to their center of masses :

(A) $\lambda_{CM} = \lambda_1 = \lambda_2$
(B) $\lambda_{CM} = \frac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2+\lambda_2^2}}$
(C) $\frac{1}{\lambda_{CM}} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$
(D) $\lambda_{CM} = \left(\frac{\lambda_1+\lambda_2}{2}\right)$

Open solution

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Q161. JEE Main 2017 (Online) 9th April Morning Slot

A Laser light of wavelength 660 nm is used to weld Retina detachment. If a Laser pulse of width 60 ms and power 0.5 kW is used the approximate number of photons in the pulse are :

Open solution

[Take Planck's constant h = 6.62×10^{-34} Js]

(A) 10^{20} (B) 10^{18} (C) 10^{22} (D) 10^{19}

Q162. JEE Main 2017 (Online) 8th April Morning Slot

The maximum velocity of the photoelectrons emitted from the surface is v when light of frequency n falls on a metal surface. If the incident frequency is increased to 3n, the maximum velocity of the ejected photoelectrons will be :

Open solution

(A) less than $\sqrt{3}$ v (B) v (C) more than $\sqrt{3}v$ (D) equal to $\sqrt{3}v$

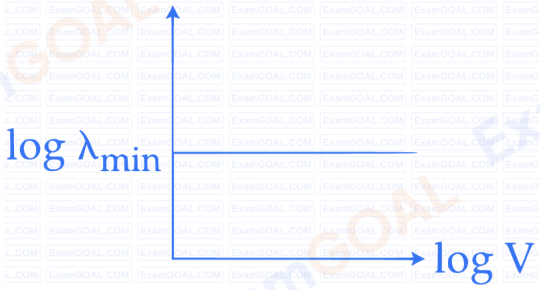
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Q163. JEE Main 2017 (Offline)

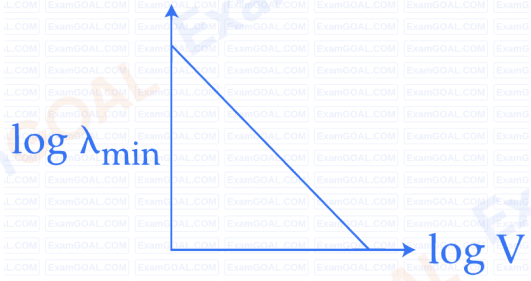
Open solution

An electron beam is accelerated by a potential difference V to hit a metallic target to produce X–rays. It produces continuous as well as characteristic X-rays. If λ_{min} is the smallest possible wavelength of X-ray in the spectrum, the variation of $\log\lambda_{min}$ with log V is correctly represented in:

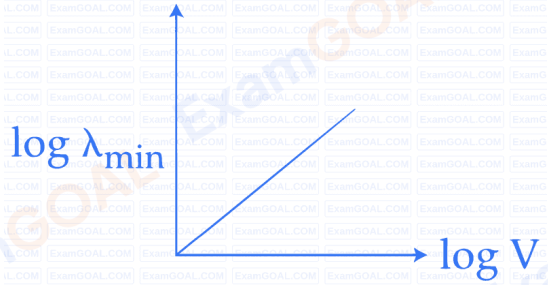
(A)



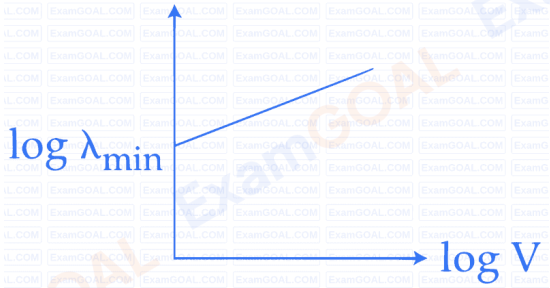
(B)



(C)



(D)



Q164. JEE Main 2017 (Offline)

Open solution

A particle A of mass m and initial velocity v collides with a particle B of mass m/2 which is at rest. The collision is head on, and elastic. The ratio of the de-Broglie wavelengths λ_A to λ_B after the collision is:

(A) $\frac{\lambda_A}{\lambda_B} = \frac{1}{3}$

(B) $\frac{\lambda_A}{\lambda_B} = 2$

(C) $\frac{\lambda_A}{\lambda_B} = \frac{2}{3}$

(D) $\frac{\lambda_A}{\lambda_B} = \frac{1}{2}$

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Q165. JEE Main 2016 (Online) 10th April Morning Slot

A photoelectric surface is illuminated successively by monochromatic light of wavelengths λ and $\frac{\lambda}{2}$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface is :

Open solution

(A) $\frac{hc}{3\lambda}$ (B) $\frac{hc}{2\lambda}$ (C) $\frac{hc}{\lambda}$ (D) $\frac{3\,hc}{\lambda}$

Q166. JEE Main 2016 (Online) 9th April Morning Slot

When photons of wavelength λ_1 are incident on an isolated sphere, the corresponding stopping potential is found to be V. When photons of wavelength λ_2 are used, the corresponding stopping potential was thrice that of the above value. If light of wavelength λ_3 is used then find the stopping potential for this case :

Open solution

(A) $\frac{hc}{e}\left[\frac{1}{\lambda_3}-\frac{1}{\lambda_2}-\frac{1}{\lambda_1}\right]$

(B) $\frac{hc}{e}\left[\frac{1}{\lambda_3}+\frac{1}{\lambda_2}-\frac{1}{\lambda_1}\right]$

(C) $\frac{hc}{e}\left[\frac{1}{\lambda_3}+\frac{1}{2\lambda_2}-\frac{3}{2\lambda_1}\right]$

(D) $\frac{hc}{e}\left[\frac{1}{\lambda_3}+\frac{1}{2\lambda_2}-\frac{1}{\lambda_1}\right]$

Q167. JEE Main 2016 (Offline)

Radiation of wavelength λ, is incident on a photocell. The fastest emitted electron has speed v. If the wavelength is changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electron will be:

Open solution

(A) $= v\left(\frac{4}{3}\right)^{\frac{1}{2}}$ (B) $= v\left(\frac{3}{4}\right)^{\frac{1}{2}}$

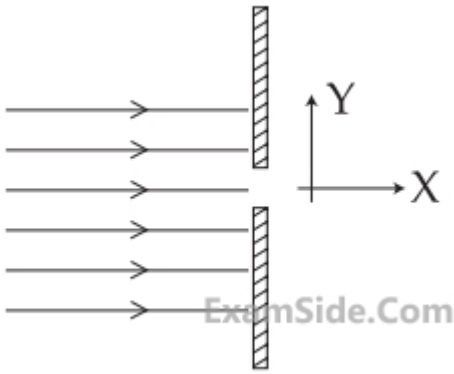
(C) $> v\left(\frac{4}{3}\right)^{\frac{1}{2}}$ (D) $< v\left(\frac{4}{3}\right)^{\frac{1}{2}}$

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Q168. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

A parallel beam of electrons travelling in x-direction falls on a slit of width d (see figure). If after passing the slit, an electron acquires momentum p_y in the y-direction then for a majority of electrons passing through the slit (h is Planck's constant) :



(A) $|p_y|d \simeq h$

(B) $|p_y|d > h$

(C) $|p_y|d < h$

(D) $|p_y|d >> h$

Q169. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

de-Broglie wavelength of an electron accelerated by a voltage of 50 V is close to ($|e|$ = 1.6 $\times$ 10^{-19} C, m_e = 9.1 $\times$ 10^{-31} kg, h = 6.6 $\times$ 10^{-34} Js) :

(A) 0.5 $\overset{o}{A}$

(B) 1.2 $\overset{o}{A}$

(C) 1.7 $\overset{o}{A}$

(D) 2.4 $\overset{o}{A}$

Q170. JEE Main 2014 (Online) 19th April Morning Slot

Open solution

Match **List - I** (Experiment performed) with **List - II** (Phenomena discovered/associated) and select the correct option from the options given below the lists :

List - I		List - II	
(a)	Davisson and Germer Experiment	(i)	Wave nature of electrons
(b)	Millikan's oil drop experiment	(ii)	Charge of an electron
(c)	Rutherford experiment	(iii)	Quantisation of energy levels
(d)	Franck - Hertz experiment	(iv)	Existence of nucleus

(A) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(B) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

(C) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(D) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

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Q171. JEE Main 2014 (Online) 12th April Morning Slot

A beam of light has two wavelengths 4972 $\mathring{A}$ and 6216 $\mathring{A}$ with a total intensity of 3.6×10^{-3} Wm^{-2} equally distributed among the two wavelengths. The beam falls normally on an area of 1 cm^2 of a clean metallic surface of work function 2.3 eV. Assume that there is no loss of light by reflection and that each capable photon ejects one electron. The number of photo electrons liberated in 2s is approximately :

Open solution

(A) 6×10^{11} (B) 9×10^{11} (C) 11×10^{11} (D) 15×10^{11}

Q172. JEE Main 2014 (Online) 11th April Morning Slot

A photon of wavelength λ is scattered from an electron, which was at rest. The wavelength shift $\Delta\lambda$ is three times of λ and the angle of scattering θ is 60°. The angle at which the electron recoiled is ϕ. The value of tan ϕ is : (electron speed is much smaller than the speed of light)

Open solution

(A) 0.16 (B) 0.22 (C) 0.25 (D) 0.28

Q173. JEE Main 2014 (Online) 9th April Morning Slot

For which of the following particles will it be most difficult to experimentally verify the de-Broglie relationship ?

Open solution

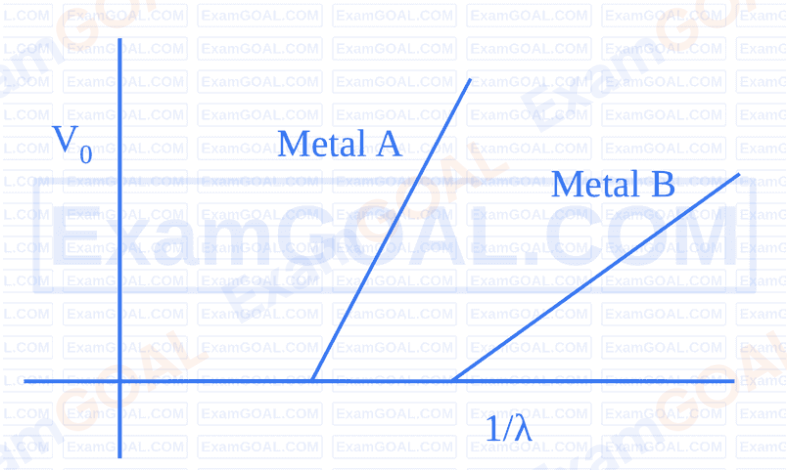
(A) an electron (B) a proton (C) an α-particle (D) a dust particle

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Q174. JEE Main 2013 (Online) 25th April Morning Slot

Open solution

In an experiment on photoelectric effect, a student plots stopping potential V_0 against reciprocal of the wavelength λ of the incident light for two different metals A and B. These are shown in the figure.



Looking at the graphs, you can most appropriately say that:

(A) Work function of metal B is greater than that of metal A

(B) For light of certain wavelength falling on both metal, maximum kinetic energy of electrons emitted from A will be greater than those emitted from B.

(C) Work function of metal A is greater than that of metal B

(D) Students data is not correct

Q175. JEE Main 2013 (Online) 25th April Morning Slot

Open solution

A copper ball of radius 1 cm and work function 4.47 eV is irradiated with ultraviolet radiation of wavelength 2500 Å. The effect of irradiation results in the emission of electrons from the ball. Further the ball will acquire charge and due to this there will be a finite value of the potential on the ball. The charge acquired by the ball is :

(A) 5.5×10^{-13} C (B) 7.5×10^{-13} C (C) 4.5×10^{-12} C (D) 2.5×10^{-11} C

Q176. JEE Main 2013 (Online) 23th April Morning Slot

Open solution

Electrons are accelerated through a potential difference V and protons are accelerated through a potential difference 4 V. The de-Broglie wavelengths are λ_e and λ_p for electrons and protons respectively. The ratio of $\frac{\lambda_e}{\lambda_p}$ is given by: (given m_e is mass of electron and m_p is mass of proton).

(A) $\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}}$ (B) $\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_e}{m_p}}$

(C) $\frac{\lambda_e}{\lambda_p} = \frac{1}{2}\sqrt{\frac{m_e}{m_p}}$ (D) $\frac{\lambda_e}{\lambda_p} = 2\sqrt{\frac{m_p}{m_e}}$

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Q177. JEE Main 2013 (Online) 22th April Morning Slot

Open solution

Orbits of a particle moving in a circle are such that the perimeter of the orbit equals an integer number of de-Broglie wavelengths of the particle. For a charged particle moving in a plane perpendicular to a magnetic field, the radius of the nth orbital will therefore be proportional to :

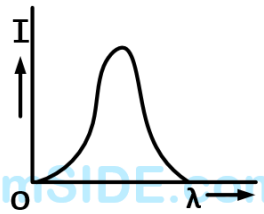
(A) n^2 (B) n (C) $n^{1/2}$ (D) $n^{1/4}$

Q178. JEE Main 2013 (Offline)

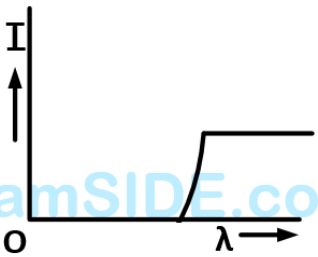
Open solution

The anode voltage of a photocell is kept fixed. The wavelength λ of the light falling on the cathode is gradually changed. The plate current I of the photocell varies as follows :

(A)



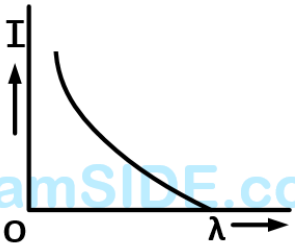
(B)



(C)



(D)



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Q179. AIEEE 2012 (Online) 26th May

Photoelectrons are ejected from a metal when light of frequency ν falls on it. Pick out the wrong statement from the following.

(A) No electrons are emitted if ν is less than W/h, where W is the work function of the metal

(B) The ejection of the photoelectrons is instantaneous.

(C) The maximum energy of the photoelectrons is $h\nu$.

(D) The maximum energy of the photoelectrons is independent of the intensity of the light.

Open solution

Q180. AIEEE 2012 (Online) 19th May

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: A metallic surface is irradiated by a monochromatic light of frequency $v > v_0$ (the threshold frequency). If the incident frequency is now doubled, the photocurrent and the maximum kinetic energy are also doubled.
Statement 2: The maximum kinetic energy of photoelectrons emitted from a surface is linearly dependent on the frequency of the incident light. The photocurrent depends only on the intensity of the incident light.

(A) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.

(B) Statement 1 is false, Statement 2 is true.

(C) Statement 1 is true, Statement 2 is false.

(D) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.

Open solution

Q181. AIEEE 2012 (Online) 12th May

A $10\,\mathrm{kW}$ transmitter emits radio waves of wavelength $500\,\mathrm{m}$. The number of photons emitted per second by the transmitter is of the order of

(A) 10^{37} (B) 10^{31} (C) 10^{25} (D) 10^{43}

Open solution

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Q182. AIEEE 2011

Open solution

This question has Statement - 1 and Statement - 2. Of the four choices given after the statements, choose the one that best describes the two statements.

Statement - 1 : A metallic surface is irradiated by a monochromatic light of frequency $v > v_0$ (the threshold frequency). The maximum kinetic energy and the stopping potential are $K_{\max}$ and V_0 respectively. If the frequency incident on the surface is doubled, both the $K_{\max}$ and V_0 are also doubled.

Statement - 2 : The maximum kinetic energy and the stopping potential of photoelectrons emitted from a surface are linearly dependent on the frequency of incident light.

(A) Statement - 1 is true, Statement - 2 is true, Statement - 2 is the correct explanation of Statement - 1.

(B) Statement - 1 is true, Statement - 2 is true, Statement - 2 is not the correct explanation of Statement - 1.

(C) Statement - 1 is false, Statement - 2 is true.

(D) Statement - 1 is true, Statement - 2 is false.

Q183. AIEEE 2010

Open solution

Statement - 1 : When ultraviolet light is incident on a photocell, its stopping potential is V_0 and the maximum kinetic energy of the photoelectrons is $K_{\max}$. When the ultraviolet light is replaced by X-rays, both V_0 and $K_{\max}$ increase.

Statement - 2 : Photoelectrons are emitted with speeds ranging from zero to a maximum value because of the range of frequencies present in the incident light.

(A) Statement - 1 is true, Statement - 2 is true; Statement - 2 is the correct explanation of Statement - 1

(B) Statement - 1 is true, Statement - 2 is true; Statement - 2 is not the correct explanation of Statement - 1

(C) Statement - 1 is false, Statement - 2 is true

(D) Statement - 1 is true, Statement - 2 is false

Q184. AIEEE 2010

Open solution

If a source of power $4kW$ produces 10^{20} photons/second, the radiation belongs to a part of the spectrum called

(A) X -rays (B) ultraviolet rays (C) microwaves (D) γ - rays

Q185. AIEEE 2009

Open solution

The surface of a metal is illuminated with the light of $400\ nm$. The kinetic energy of the ejected photoelectrons was found to be $1.68\ eV$. The work function of the metal is : $(hc = 1240eV.nm)$

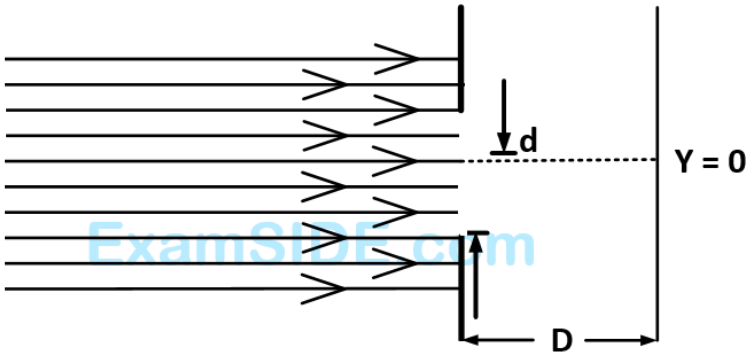
(A) $1.41\ eV$ (B) $1.51\ eV$ (C) $1.68\ eV$ (D) $3.09\ eV$

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Q186. AIEEE 2008

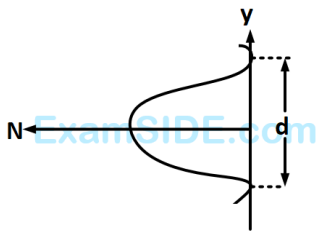
Open solution

In an experiment, electrons are made to pass through a narrow slit of width $'d'$ comparable to their de Broglie wavelength. They are detected on a screen at a distance $'D'$ from the slit (see figure).

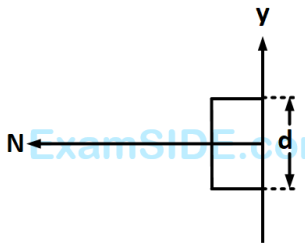


Which of the following graphs can be expected to represent the number of electrons $'N'$ detected as a function of the detector position $'y'$ $(y = 0$ corresponds to the middle of the slit $)$

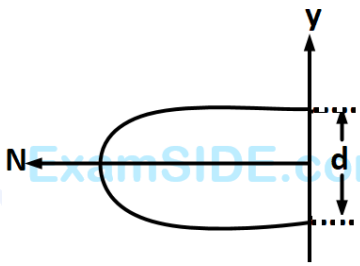
(A)



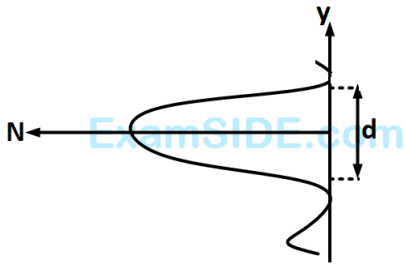
(B)



(C)



(D)



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Q187. AIEEE 2007

Photon of frequency v has a momentum associated with it. If c is the velocity of light, the momentum is

(A) hv/c (B) v/c (C) $h\,v\,c$ (D) hv/c^2

Open solution

Q188. AIEEE 2006

The threshold frequency for a metallic surface corresponds to an energy of $6.2\,eV$ and the stopping potential for a radiation incident on this surface is $5V$. The incident radiation lies in

(A) ultra-violet region (B) infra-red region

(C) visible region (D) x-ray region

Open solution

Q189. AIEEE 2006

The time taken by a photoelectron to come out after the photon strikes is approximately

(A) $10^{-4}\,s$ (B) $10^{-10}\,s$ (C) $10^{-16}\,s$ (D) $10^{-1}\,s$

Open solution

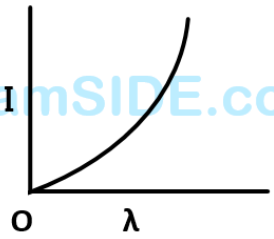
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Q190. AIEEE 2006

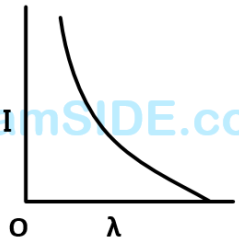
The anode voltage of a photocell is kept fixed. The wavelength λ of the light falling on the cathode is gradually changed. The plate current I of the photocell varies as follows

Open solution

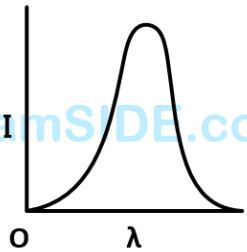
(A)



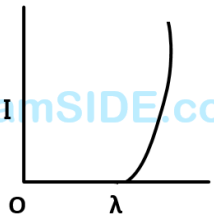
(B)



(C)



(D)



Q191. AIEEE 2005

A photocell is illuminated by a small bright source placed $1\ m$ away. When the same source of light is placed $\frac{1}{2}\ m$ away, the number of electrons emitted by photo-cathode would

Open solution

(A) increases by a factor of 4

(B) decreases by a factor of 4

(C) increases by a factor of 2

(D) decreases by a factor of 2

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Q192. AIEEE 2005

If the kinetic energy of a free electron doubles, it's deBroglie wavelength changes by the factor

(A) 2 (B) $\frac{1}{2}$ (C) $\sqrt{2}$ (D) $\frac{1}{\sqrt{2}}$

Open solution

Q193. AIEEE 2004

According to Einstein's photoelectric equation, the plot of the kinetic energy of the emitted photo electrons from a metal Vs the frequency of the incident radiation gives a straight line whose slope

(A) depends both on the intensity of the radiation and the metal used

(B) depends on the intensity of the radiation

(C) depends on the nature of the metal used

(D) is the same for all metals and independent of the intensity of the radiation

Open solution

Q194. AIEEE 2004

The work function of a substance is $4.0\,eV$. The longest wavelength of light that can cause photo-electron emission from this substance is approximately.

(A) $310\,nm$ (B) $400\,nm$ (C) $540\,nm$ (D) $220\,nm$

Open solution

Q195. AIEEE 2003

Two identical photo-cathodes receive light of frequencies f_1 and f_2. If the velocities of the photo electrons (of mass m) coming out are respectively v_1 and v_2, then

(A) $v_1^2 - v_2^2 = \frac{2h}{m}(f_1 - f_2)$

(B) $v_1 + v_2 = \left[\frac{2h}{m}(f_1 + f_2)\right]^{1/2}$

(C) $v_1^2 + v_2^2 = \frac{2h}{m}(f_1 + f_2)$

(D) $v_1 - v_2 = \left[\frac{2h}{m}(f_1 - f_2)\right]^{1/2}$

Open solution

Q196. AIEEE 2002

Sodium and copper have work functions $2.3\,eV$ and $4.5\,eV$ respectively. Then the ratio of the wavelengths is nearest to

(A) $1:2$ (B) $4:1$ (C) $2:1$ (D) $1:4$

Open solution

Q197. AIEEE 2002

Formation of covalent bonds in compounds exhibits

(A) wave nature of electron

(B) particle nature of electron

(C) both wave and particle nature of electron

(D) none of these

Open solution

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Numerical 16 questions

Q198. JEE Main 2026 (Online) 6th April Evening Shift

The de Broglie wavelength for an electron accelerated through the potential difference of V_1 volt is λ_1. When the potential difference is changed to V_2 volt, the associated de Broglie wavelength is increased by 50%. If $(V_1/V_2) = (9/\alpha)$, then the value of α is ____.

Open solution

Q199. JEE Main 2026 (Online) 28th January Morning Shift

The ratio of de Broglie wavelength of a deuteron with kinetic energy E to that of an alpha particle with kinetic energy $2E$, is $n : 1$. The value of n is ____ .

(Assume mass of proton = mass of neutron)

Open solution

Q200. JEE Main 2026 (Online) 21st January Evening Shift

A particle having electric charge $3 \times 10^{-19}\text{C}$ and mass 6×10^{-27} kg is accelerated by applying an electric potential of 1.21 V .

Wavelength of the matter wave associated with the particle is $\alpha \times 10^{-12}$ m. The value of α is ____ .

(Take Planck's constant $= 6.6 \times 10^{-34}$ J. s)

Open solution

Q201. JEE Main 2025 (Online) 8th April Evening Shift

An electron is released from rest near an infinite non-conducting sheet of uniform charge density '$-\sigma$'. The rate of change of de-Broglie wave length associated with the electron varies inversely as n^{th} power of time. The numerical value of n is ____.

Open solution

Q202. JEE Main 2025 (Online) 24th January Evening Shift

The ratio of the power of a light source S_1 to that of the light source S_2 is 2. S_1 is emitting 2×10^{15} photons per second at 600 nm . If the wavelength of the source S_2 is 300 nm , then the number of photons per second emitted by S_2 is _____ $\times 10^{14}$.

Open solution

Q203. JEE Main 2023 (Online) 13th April Evening Shift

An atom absorbs a photon of wavelength 500 nm and emits another photon of wavelength 600 nm. The net energy absorbed by the atom in this process is $n \times 10^{-4}$ eV. The value of n is _____. [Assume the atom to be stationary during the absorption and emission process] (Take $h = 6.6 \times 10^{-34}$ Js and $c = 3 \times 10^8$ m/s)

Open solution

Q204. JEE Main 2022 (Online) 26th June Evening Shift

The stopping potential for photoelectrons emitted from a surface illuminated by light of wavelength 6630 $\mathring{A}$ is 0.42 V. If the threshold frequency is x $\times$ 10^{13} /s, where x is _____ (nearest integer).

(Given, speed light = 3 $\times$ 10^8 m/s, Planck's constant = 6.63 $\times$ 10^{-34} Js)

Open solution

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Q205. JEE Main 2022 (Online) 24th June Morning Shift

When light of frequency twice the threshold frequency is incident on the metal plate, the maximum velocity of emitted electron is v_1. When the frequency of incident radiation is increased to five times the threshold value, the maximum velocity of emitted electron becomes v_2. If $v_2 = x\ v_1$, the value of x will be _____.

Open solution

Q206. JEE Main 2021 (Online) 27th July Morning Shift

A particle of mass 9.1×10^{-31} kg travels in a medium with a speed of 10^6 m/s and a photon of a radiation of linear momentum 10^{-27} kg m/s travels in vacuum. The wavelength of photon is _____ times the wavelength of the particle.

Open solution

Q207. JEE Main 2021 (Online) 25th July Evening Shift

A light beam of wavelength 500 nm is incident on a metal having work function of 1.25 eV, placed in a magnetic field of intensity B. The electrons emitted perpendicular to the magnetic field B, with maximum kinetic energy are bent into circular arc of radius 30 cm. The value of B is _____ $\times$ 10^{-7} T.

Given hc = 20×10^{-26} J-m, mass of electron = 9×10^{-31} kg

Open solution

Q208. JEE Main 2021 (Online) 20th July Evening Shift

A certain metallic surface is illuminated by monochromatic radiation of wavelength λ. The stopping potential for photoelectric current for this radiation is $3V_0$. If the same surface is illuminated with a radiation of wavelength 2λ, the stopping potential is V_0. The threshold wavelength of this surface for photoelectric effect is _____ λ.

Open solution

Q209. JEE Main 2021 (Online) 26th February Evening Shift

Two streams of photons, possessing energies equal to twice and ten times the work function of metal are incident on the metal surface successively. The value of ratio of maximum velocities of the photoelectrons emitted in the two respective cases is x : y. The value of x is _____.

Open solution

Q210. JEE Main 2021 (Online) 25th February Evening Shift

The wavelength of an X-ray beam is $10\overset{o}{A}$. The mass of a fictitious particle having the same energy as that of the X-ray photons is $\frac{x}{3}h$ kg. The value of x is _____. (h = Planck's constant)

Open solution

Q211. JEE Main 2020 (Online) 5th September Evening Slot

The surface of a metal is illuminated alternately with photons of energies E_1 = 4 eV and E_2 = 2.5 eV respectively. The ratio of maximum speeds of the photoelectrons emitted in the two cases is 2. The work function of the metal in (eV) is _____.

Open solution

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Q212. JEE Main 2020 (Online) 2nd September Morning Slot

When radiation of wavelength λ is used to illuminate a metallic surface, the stopping potential is V. When the same surface is illuminated with radiation of wavelength 3λ, the stopping potential is $\frac{V}{4}$. If the threshold wavelength for the metallic surface is $n\lambda$ then value of n will be _____.

Open solution

Q213. JEE Main 2020 (Online) 7th January Morning Slot

A beam of electromagnetic radiation of intensity 6.4×10^{-5} W/cm^2 is comprised of wavelength, λ = 310 nm. It falls normally on a metal (work function ϕ = 2eV) of surface area of 1 cm^2. If one in 10^3 photons ejects an electron, total number of electrons ejected in 1 s is 10^x. (hc = 1240 eVnm, 1eV = 1.6×10^{-19} J), then x is _____.

Open solution

Out of syllabus 8 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q214. JEE Main 2023 (Online) 31st January Morning Shift · MCQ (Single Correct Answer)

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R

Assertion A : The beam of electrons show wave nature and exhibit interference and diffraction.

Reason R : Davisson Germer Experimentally verified the wave nature of electrons.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) A is not correct but R is correct.

(B) Both A and R are correct and R is the correct explanation of A

(C) Both A and R are correct but R is Not the correct explanation of A

(D) A is correct but R is not correct

Open solution

Q215. JEE Main 2022 (Online) 25th June Morning Shift · MCQ (Single Correct Answer)

Given below are two statements :

Statement I : Davisson-Germer experiment establishes the wave nature of electrons.

Statement II : If electrons have wave nature, they can interfere and show diffraction.

In the light of the above statements choose the correct answer from the option given below :

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Open solution

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Q216. JEE Main 2021 (Online) 18th March Evening Shift · MCQ (Single Correct Answer)

Open solution

The speed of electrons in a scanning electron microscope is 1×10^7 ms^{-1}. If the protons having the same speed are used instead of electrons, then the resolving power of scanning proton microscope will be changed by a factor of :

(A) $\frac{1}{1837}$ (B) 1837 (C) $\sqrt{1837}$ (D) $\frac{1}{\sqrt{1837}}$

Q217. JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

Open solution

Match **List -** I (Fundamental Experiment) with **List -** II (its conclusion) and select the correct option from the choices given below the list:

List - I		List - II	
A.	Franck-Hertz	(i)	Particle nature of light
B.	Photo-electric	(ii)	Discrete energy levels of atom
C.	Davison-Germer experiment	(iii)	Wave nature of electron
		(iv)	Structure of atom

(A) $A-ii;\ B-i,\ C-iii$ (B) $A-iv;\ B-iii,\ C-ii$

(C) $A-i;\ B-iv,\ C-iii$ (D) $A-ii;\ B-iv,\ C-iii$

Q218. AIEEE 2012 · MCQ (Single Correct Answer)

Open solution

This question has Statement 1 and Statement 2. Of the four choices given after the statements, choose the one that best describes the two statements.

Statement 1 : Davisson - Germer experiment established the wave nature of electrons.

Statement 2 : If electrons have wave nature, they can interfere and show diffraction.

(A) Statement 1 is true, Statement 2 is false

(B) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation for Statement 1.

(C) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.

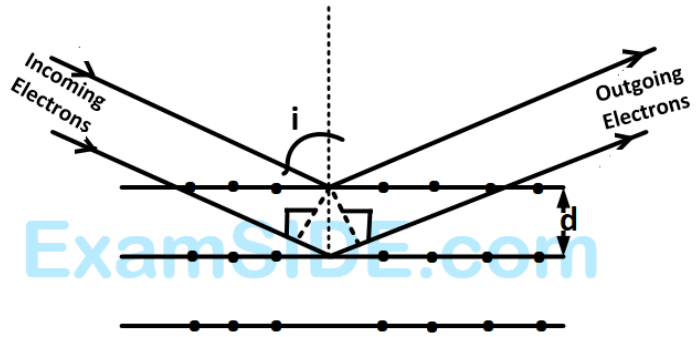
(D) Statement 1 is false, Statement 2 is true.

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Q219. AIEEE 2008 · MCQ (Single Correct Answer)

Open solution

Wave property of electrons implies that they will show diffraction effects. Davisson and Germer demonstrated this by diffracting electrons from crystals. The law governing the diffraction from a crystal is obtained by requiring that electron waves reflected from the planes of atoms in a crystal interfere constructively (see figure).



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Electrons accelerated by potential V are diffracted from a crystal. If $d = 1\mathring{A}$ and $i = 30^{\circ}$, V should be about

$(h = 6.6 \times 10^{-34} Js, m_e = 9.1 \times 10^{-31} kg, e = 1.6 \times 10^{-19} C)$

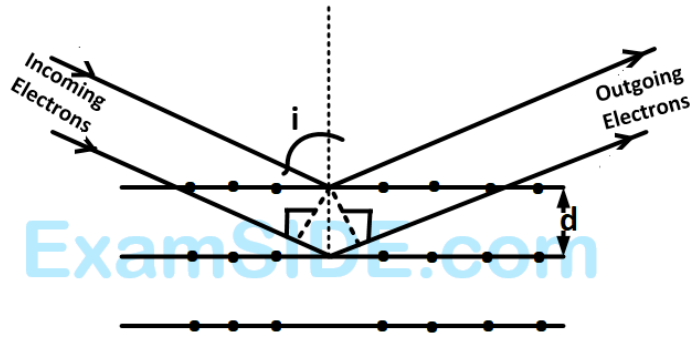
(A) $2000\ V$ (B) $50\ V$ (C) $500\ V$ (D) $1000\ V$

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Q220. AIEEE 2008 · MCQ (Single Correct Answer)

Open solution

Wave property of electrons implies that they will show diffraction effects. Davisson and Germer demonstrated this by diffracting electrons from crystals. The law governing the diffraction from a crystal is obtained by requiring that electron waves reflected from the planes of atoms in a crystal interfere constructively (see figure).



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If a strong diffraction peak is observed when electrons are incident at an angle $'i'$ from the normal to the crystal planes with distance $'d'$ between them (see figure), de Broglie wavelength λ_{dB} of electrons can be calculated by the relationship (n is an integer)

(A) $d \sin i = n\lambda_{dB}$ (B) $2d \cos i = n\lambda_{dB}$ (C) $2d \sin i = n\lambda_{dB}$ (D) $d \cos i = n\lambda_{dB}$

Q221. JEE Main 2020 (Online) 5th September Morning Slot · Numerical

Open solution

A beam of electrons of energy E scatters from a target having atomic spacing of 1 $\overset{o}{A}$. The first maximum intensity occurs at θ = 60°. Then E (in eV) is _____.
(Planck constant h = 6.64 × 10^{-34} Js,
1 eV = 1.6 × 10^{-19} J, electron
mass m = 9.1 × 10^{-31} kg)

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Answer Key

MCQ (Single Correct Answer)

1. **D**	2. **D**	3. **C**	4. **C**	5. **C**
6. **A**	7. **A**	8. **B**	9. **D**	10. **B**
11. **D**	12. **D**	13. **D**	14. **C**	15. **D**
16. **B**	17. **C**	18. **D**	19. **B**	20. **B**
21. **A**	22. **D**	23. **B**	24. **D**	25. **D**
26. **A**	27. **C**	28. **C**	29. **A**	30. **A**
31. **D**	32. **A**	33. **B**	34. **B**	35. **C**
36. **C**	37. **B**	38. **A**	39. **A**	40. **C**
41. **A**	42. **A**	43. **C**	44. **A**	45. **A**
46. **C**	47. **B**	48. **A**	49. **A**	50. **C**
51. **B**	52. **C**	53. **C**	54. **B**	55. **B**
56. **B**	57. **A**	58. **A**	59. **A**	60. **D**
61. **C**	62. **D**	63. **A**	64. **C**	65. **A**
66. **A**	67. **C**	68. **C**	69. **A**	70. **D**
71. **D**	72. **C**	73. **C**	74. **C**	75. **C**
76. **D**	77. **D**	78. **A**	79. **B**	80. **B**
81. **C**	82. **D**	83. **B**	84. **D**	85. **C**
86. **B**	87. **D**	88. **C**	89. **A**	90. **D**
91. **B**	92. **B**	93. **A**	94. **B**	95. **A**
96. **B**	97. **B**	98. **B**	99. **A**	100. **C**
101. **C**	102. **A**	103. **B**	104. **D**	105. **B**
106. **A**	107. **B**	108. **C**	109. **B**	110. **C**
111. **D**	112. **D**	113. **C**	114. **C**	115. **D**
116. **D**	117. **B**	118. **D**	119. **B**	120. **D**
121. **C**	122. **B**	123. **D**	124. **B**	125. **C**
126. **C**	127. **D**	128. **D**	129. **C**	130. **B**
131. **D**	132. **B**	133. **A**	134. **C**	135. **D**
136. **C**	137. **B**	138. **C**	139. **C**	140. **B**
141. **B**	142. **D**	143. **A**	144. **D**	145. **B**
146. **B**	147. **D**	148. **B**	149. **B**	150. **C**


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151. **C**	152. **B**	153. **B**	154. **C**	155. **A**
156. **D**	157. **A**	158. **B**	159. **A**	160. **B**
161. **A**	162. **C**	163. **B**	164. **B**	165. **B**
166. **C**	167. **C**	168. **C**	169. **C**	170. **B**
171. **B**	172. **C**	173. **D**	174. **D**	175. **A**
176. **D**	177. **C**	178. **D**	179. **C**	180. **B**
181. **B**	182. **C**	183. **D**	184. **A**	185. **A**
186. **D**	187. **A**	188. **A**	189. **B**	190. **B**
191. **A**	192. **D**	193. **D**	194. **A**	195. **A**
196. **C**	197. **A**			

Numerical

198. **4**	199. **2**	200. **10**	201. **2**	202. **5**
203. **4125**	204. **35**	205. **2**	206. **910**	207. **125**
208. **4**	209. **1**	210. **10**	211. **2**	212. **9**
213. **11**				

Out of syllabus

214. **B**	215. **A**	216. **B**	217. **A**	218. **B**
219. **B**	220. **B**	221. **50.47**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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